TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

A Research
Presented to the Faculty of
College of Computer Studies
Camarines Sur Polytechnic College

In partial fulfillment of the Requirements
For the degree of BACHELOR OF LIBRARY AND INFORMATION SCIENCE

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June 2024

DEDICATION

The study is entirely devoted to our dear parents, who have encouraged motivation and strength when we felt like giving up and continue to guide us in all areas of life.

To our family members, instructors, companions, and classmates who encouraged us to complete our research.

Finally, we dedicated our work to almighty God, thanking him for his guidance, courage, mental influence, assurance, and skills and for providing us with a life of wellness. We offer you with all of these.

BLACK MANTA

ACKNOWLEDGEMENT

We express our heartfelt gratitude and admiration to all of the people who have selflessly contributed their assistance and encouragement in making this study work a reality; we also express our sincere appreciation and affection to the following individuals who have made this accomplishment attainable:

To **Dr. Challiz D. Omorog,** Dean of the College of Computer Studies, for her appealing yet motivating encouragement to complete this study.

To **Ms. Kay Angeline O. Broqueza,** our subject adviser, thank you for your support throughout our study, your compassion, patience, and understanding, and for giving us the expertise we needed to complete the study.

To **Mr. James Nicolo SJ. Sias,** our Thesis Adviser, for his kindness, recommendations, and fantastic encouragement to finish the study, and for the inspiration he gave us to finish the research as well as for supporting us with what we needed;

To the panelists, **Ronabeth L. Fider, Jean Maricon Regaspi,** and **Mildred A. Volante,** for their informative critiques, which led to the improvement of our study.

To **Engr. Rowel Domingo N. Bermal,** for his benign and selfless statistical advice, guided the researcher in managing the data.

To **Ms. Elaine Mae G. Tormes,** our research editor, for her helpful recommendations in revising the paper and for her super grammatical revisions that enabled us to complete this research.

To all of our classmates, friends, and loved ones for their encouraging and persuasive messages, ideas, and prayers, as well as for the fantastic companionship, assistance, support, and encouragement that gave the researchers more courage and motivation while we worked on this study;

To our parents and family for their moral and financial assistance, as well as their unending contributions during the completion of this study;

To the respondents for their involvement in answering the survey questionnaire, which allowed the researchers to collect the information they needed;

To individuals who were not mentioned but contributed significantly to this academic endeavor by sharing their valuable time,

Above everything, to the HEAVENLY FATHER, who makes everything possible and without whom we are nothing.

The researchers appreciate their support and are grateful to every one of them.

ABSTRACT

Title	:	Technology Integration Promoting Reading Literacy in Iriga City Public Library
Authors	:	Jhoemie A. Ya-on Beverly G. Jacob Ma.Therese T.Sombrero
Number of Pages	:	119 pages
School	:	Camarines Sur Polytechnic Colleges
Degree Conferred	:	Bachelor of Library Information Science
Keywords	:	Technologies, Technology integration, Reading Literacy, Iriga City, Public Library

The study explores integrating technology to enhance reading literacy at the Iriga City Public Library. It aims to identify specific technologies that can improve reading literacy among library patrons while also addressing the challenges of incorporating digital tools in a traditional library setting in the Iriga City Public Library. The researchers employed a quantitative, descriptive design in gathering the data and used the Technology Acceptance Model to examine technology adoption and user behavior. The study involved 355 respondents, including students, professionals, parents, and others such as PWDs, pregnant women, senior citizens, and out-of-school youth, from the library user's login monthly. The research study findings revealed that most respondents were students, indicating that the library is frequently used for academic purposes due to its proximity to schools and the accessibility of information. Most students preferred electronic gadgets, which was reflected as strongly agreeing, firmly believing, or very optimistic about integrating them into the public library for reading. However, the study also highlighted various challenges, such as the accessibility of digital tools that will be implemented in the public library. To address these challenges, the researchers proposed solutions which is the contingency plan focusing on enhancing user experience, identifying risks, providing solutions, having backup plans, and promoting adaptability and resilience. The overall goal is to ensure continuous service, protect resources, and improve the reading literacy of library patrons through effective technology integration. This approach aims to build trust, enhance user experience, and ensure the sustainability of technology use in the library.

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Chapter 1

INTRODUCTION

Background of the study

Libraries are the ocean of knowledge and essential educational, research, and information resources. Information and communication technology (ICT) is advancing to the point where educational institutions are turning their digital into mobile library applications. This is especially true of wireless and mobile technologies. A Library accessed by a user through a computer and web broadband at the defined place is the definition of a digital library. Users can access library resources from the library, their home, workplace, or an online library by using a computer at a designated location and network. Additionally, users of digital libraries do general searches using distinct search engines compared to those used in traditional libraries. It illustrates how time and space are limited for conventional and digital libraries, so accessing library services requires a specific location, system, and internet [1].

The Public Library is a building for society. The members of the community created and oversaw this. The public library is an invaluable resource for people's lives. This organization is beneficial to people and plays a significant part in the advancement of society. Public libraries have a crucial social function and role in ensuring that the lifestyles of all societal segments are wellsuited. It was their responsibility to raise public knowledge of libraries and public libraries. The reason public libraries are vital is that they impart knowledge to people. Inspires everyone, everywhere. It brings out a person's untapped abilities and gives them opportunities. We share our cultural and social legacy with the globe—search material for reasonably priced goods [2].

The precise design of the technology and the pedagogical strategy determine how effective it is to teach reading comprehension. Research suggests that well-designed Digital books perform better than reading on paper. Furthermore, student-centered technology-based activities are linked to improved reading performance rather than supporting reading habits. Proactive policy actions ought to assist educators and schools in implementing student-centered technology integration strategies [3].

The employment of technological instruments in several subject areas, such as technology integration. Instruction to enable users to use their knowledge of computers and technology to learn and solve issues. It is, in essence, the application of technology to improve and facilitate the educational surroundings. Utilizing technology resources, such as computers, mobile devices like smartphones and tablets, digital cameras, social media in the ordinary classroom or library, media platforms and networks, software programs, and the Internet, etc. procedures, as well as in school administration [4].

Iriga City, Camarines Sur, a city with a population of 114,457 in the Bicol region, is home to a public library that serves a significant portion of the community. Despite the library's importance, its current condition challenges integrating new technologies for promoting reading literacy. One major obstacle is limited funding, hindering the library's ability to acquire the necessary technology. This study investigates the challenges and strategies associated with technology integration in promoting reading literacy within the Iriga City Public Library [5].

This study aims to offer insightful information to develop strategies to maximize technology integration to improve reading literacy outcomes in libraries. Through similar initiatives in other library settings, these findings can support the Iriga City Public Library to become a leader in promoting inclusive literacy in the digital age and ultimately assist in creating a more empowered and literate community.

Statement of the Problem

The researchers seek to determine the technology integration that Promotes reading literacy in the Iriga City Public Library. This will seek answers to the following questions:

1. What is the profile of the respondents: in terms of,

    3.1. Age;

3.2. Gender; and

3.3. Type of Library Users?

2. What are the technologies that can be integrated with promoting reading literacy in the Iriga City Public Library?

   2.1. Electronic Books;

   2.2. Audiobooks;

   2.3. Electronic Gadgets;

   2.4. Educational E-Games;

   2.5. Reading Applications;

   2.6. Interactive Displays;

   2.7. Online Tutoring Platform for Reading and

   2.8. ICT Facility?

3. What are the challenges encountered by Iriga City Public Library in technology integration promoting reading literacy? in terms of;

   3.1. Limited ICT Skills;

   3.2. Limited Fund for Technology Integration;

   3.3. Library Space and Capacity;

   3.4. Internet Connectivity;

   3.5. Usage Policies;

   3.6. Lack of IT and staff;

   3.7. Technology Safety; and

   3.8. Technology Accessibility?

4. What strategies can be proposed to enhance technology integration in promoting reading literacy?

Objectives

The main goal of this study is to assess the current state of the technology integration in promoting reading literacy in the Iriga City Public Library and develop a strategy to enhance and improve children's reading skills.

1. To determine the respondents' profiles, which vary in age, gender, and types of library users.
2. To identify the technology that will integrate in promoting reading literacy in Iriga City Public Library, such as e-books and audiobooks, electronic Gadgets, educational e-games, reading applications, interactive displays, online tutoring platform for reading, and ICT facility.
3. To determine the challenges Iriga City Public Library encountered in integrating technology in promoting reading literacy, such as limited funds for technology integration, library space and capacity, Internet connectivity, usage Policies, lack of IT and staff, Technology safety, and technology Accessibility.
4. To develop a strategy that can be proposed to enhance the integrated technology in promoting reading literacy.

Significance

The result of this study will benefit the following:

Iriga City Public Library. They can offer more digitalized and expanded library technology materials to their patrons.

Public Librarian. This research study will offer public librarians' helpful information on effectively integrating technology into library programs. It will give them practical advice on choosing the right digital tools, creating exciting activities, and assessing the influence of technology on the reading skills of their patrons. By applying the research study's findings, public librarians can also improve their professional skills, enhance library services, and better serve patrons in the digital age. To ensure public libraries' ongoing relevance and community accessibility, the study might be helpful to public librarians in arguing for increased financing and resources for technological integration in public libraries.

People of Iriga City. This research could lead to the implementation of new technologies that would make it easier for Iriga city residents to access books and other resources from the public library, particularly for those with limited mobility or who live in remote areas.

Educators. It also enables educators to utilize various multimedia tools to customize their lessons and cater to diverse learning preferences.

Students. Library-integrated technologies will be helpful as they will provide and assess how to use the technology to enhance reading, teaching, learning, and research.

Parents. Through this research, they can assess and evaluate high-quality educational resources and content that can support their children's reading development while teaching them about responsible digital behavior.

LGU Iriga. In this study, the integrated library technology can potentially improve Iriga City government employees regarding technology services such as speed, reliability, accuracy, convenience, and even program outcomes.

Literacy Advocacy Groups. This study can use this as a support and advocate for reading with the intervention of technologies in the public library in Iriga City. It will offer them valuable insights on using technology to improve reading literacy, promote a reading culture, and ensure that these institutions remain essential for community education and personal development.

Researchers. As a researcher, studying technology integration and reading literacy in the Iriga City Public Library can grant valuable insights, contribute to knowledge on libraries in developing countries, and potentially inform best practices for broader impact.

Future Researchers. The study's results will guide future researchers seeking relevant information for a more extensive and in-depth range of studies. As a result, it will be one of their references or guides for their study that will replicate and provide them with a deeper understanding of their studies.

Scope and Limitations

The research study focuses on the Iriga City Public Library, Rizal St, Iriga City, Camarines Sur. This study will examine a variety of technologies used, such as electronic books, audiobooks, electronic gadgets, educational e-games, reading applications, interactive displays, online tutoring platforms for reading, and ICT facilities, and identify the challenges and strategies that can be proposed to enhance the integration of technology in promoting reading literacy. Using 355 community members as a sample, this study looks into how technology integration affects the reading literacy of library users.

This study was limited and restricted to the Iriga City Public Library. The shortcomings will be determined based on the variables we use as the researcher. Time, finances, resources, data, or involvement are among the variables that could be considered. For example, if we were limited to using the convenience sampling method and could not obtain a random sample of participants, our ability to generalize the findings would be affected, indicating a constraint in the study.

Project Dictionary

The following terms were conceptually and operationally defined to understand the term used in this study.

Audio-Books. This pertains to books with pictures and a spoken presentation of the stories. It includes vocabulary gains and enhancement of listening fluency and comprehension and reading fluency improvement [6]. It is defined as a spoken version of a book published electronically.

Electronic Books. A digital file containing a body of text and images suitable for distributing electronically and displaying on-screen like a printed book [7]. As researchers, we define electronic books as the digital versions of traditional print books that offer a portable and accessible way to access and read text on electronic devices.

Interactive Displays. A digital screen that enables users to engage with content directly through touch, gestures, or other input methods. While they can be used for reading, they are primarily designed for interactive applications such as presentations, games, and collaborative activities [8]. In this study, it is defined as large digital screens provide users with a dynamic way to engage with text and information.

Public Library. Defined as a local center of information or creator of community and a provider of services to all regardless of any characteristics such as age, ethnicity, gender, religion, nationality, or social status [9]. In this research, we see a library supported by public funds, overseen by a board for the public good, open to all community members, serving as a local hub for knowledge, and offering resources and services for individuals of all ages.

Reading Literacy. A necessity in the teaching and learning process. Mastering speaking skills, such as writing and listening, is crucial [10]. This study focuses on comprehending and utilizing written language forms necessary for society or essential to the individual. Readers can derive understanding from texts in various formats.

Reading Applications. Refers to a digital tool that lets users access and read various books anytime, offering free and paid content. The installed technologies can provide new knowledge and information readily available to its user community and bookmarks [11]. As researchers, we define reading applications as software programs or applications for reading that can be available to access any device to enhance the reading literacy of the users.

Technology Integration. Refers to the strategic use of digital tools and resources to enhance the reading experience and improve literacy skills. By incorporating technology, educators and learners can access a broader range of texts, engage with content in new ways, and develop essential skills [12]. This study refers to incorporating technological tools and resources in the library to assist with reading literacy.

Technologies. The practical application of scientific knowledge, particularly in industry. It refers to applying systems and techniques to improve teaching and learning in education [13]. In this study, technologies refer to tools and platforms that boost the reading experience and enhance literacy skills. It includes e-readers, reading apps, assistive technologies, interactive whiteboards, and online libraries.

Technology Safety. Refers to the responsible and secure use of digital devices and platforms for reading activities. It encompasses practices that protect individuals from online threats, maintain privacy, and ensure a positive reading experience [14]. This study pertains to preventing injuries, accidents, or harm caused by using devices, systems, or infrastructure. This includes assessing risks, following safety protocols, maintaining equipment, and educating users about safe and beneficial technology use.

Technology Accessibility. Accessing learning materials and educational resources through mobile devices has enabled self-study in distance learning. Students can study at any place and time, enhancing the flexibility of their learning experience [15]. This study refers to the accessibility and functionality of digital tools and platforms that empower users to comprehend and interpret text efficiently. Functionalities like text-to-speech, personalized font sizes, screen readers, and audiobooks are included. This will increase the availability of reading materials for a wider audience.

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Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents selected literature and domestic and foreign studies relevant to this study. As researchers, we read several positions of the authorities in various books, magazines, and publications. Different opinions, insights, and information about the research field were collected in multiple ways.

Impacts of Technology Integration on Reading Literacy Skills in Library Users

Lou Yu Zhou, Xiao Yue Ming, Ma Yu Yang, and Li Chao [2021] mentioned that with the information era rapidly advancing, e-books' significance became apparent. The mobility of e-books has been appealing from the beginning. There was a lot of discussion about e-books in the library and information science field. Some believe that e-books should do more than be available in libraries; they should also improve library services and assist in teaching [1].

In the study of Muntaha Ali Mohammad Al Momani [2021], since smartphones and other smart gadgets have become more common, these mobile communication devices have become a necessary component of our everyday lives. The study intends to talk about the impact of contemporary technology applications to enhance kids' learning, such as telegram messenger channels to enhance reading technological aids, m-learning, and e-learning, which could potentially bridge the skills gap. EFL students are the subjects of the study at the University College of Ajloun. [2].

In the study of Wen-Chi Hu and Shih-Tsung Hsu [2024], To assist English as a foreign language learner in Taiwan to improve their reading abilities, this study intends to investigate how well a digital storytelling project that is in line with the sustainable development goals may be included into English sessions. Together with classroom observations, the research used a variety of assessment instruments, including beginning and end tests of English reading comprehension.

For this reason, there are clear benefits to teaching English as a foreign language that come from utilizing technology and incorporating sustainable development goals in the educational structure [3].

Carmen Lopez Escribano, Susana Valverde Montesino, and Veronica Garcia Ortega [2021] looked at studies that compared how well different things support young children's learning to read. They looked at interactive e-bThey found that well-designed interactive e-books can be as good or even better than other methods for helping young children learn word sounds and develop vocabulary.

Interactive e-books seemed to do more than just the usual things in school to teach children to read. This research suggests that e-books, incredibly interactive and well-designed, can be a great way to assist young children in reading [4].

The study by Taliesin L. Smith and Emily B. Moore [2020] tells the significance of designing technology inclusively to ensure its benefits are available to everyone. This objective is consistent with the Iriga City Public Library's aim of leveraging technology to enhance reading literacy and provide equal learning opportunities. Spoken descriptions can potentially improve reading comprehension for users with visual impairments [5].

This review of the literature of Mohd Nur Hifzhan Noordan and Melor Md Yunus [2022] investigates how information and communication technology (ICT), such as computers and educational software, can improve readers' comprehension. Its findings show how technology has been successfully used in the real world to improve reading skills. This information can be a valuable tool for our research project, assisting us in coming up with concepts and formulating the best plans for utilizing technology in the Iriga City Public Library to improve everyone's reading experience [6].

Sara S. Goek [2023] provides an extensive summary of technology conditions in American public libraries, as provided by the Public Library Assisiacations 2023 Public Library Technology survey. The main conclusions include a notable rise in laptop usage, comprehensive digital literacy initiatives, and the critical importance of government funding in addressing library technology needs; there is an examination of emerging trends, such as the integration of recreational technology and early learning devices in libraries. This trend offers improved services, enhanced accessibility, greater efficiency, broader resource availability, digital literacy education, community engagement, and reading literacy support. The article provides important insights into public libraries' crucial role in providing digital resources and access to their communities. [7].

Chin Ee Loh and Baoqi Sun [2022] discuss the concerns about the global decrease in children's and teenagers' pleasure of reading, as confirmed by national and international surveys, with cell phones and other devices frequently attributed to the reduction. The connection between teenagers' use of technology and reading habits might be more complex than commonly portrayed in the media, given the recent and rapid advancements in reading technologies, the increased adoption of learning devices during the pandemic, and the increased accessibility of e-books through Singapore's public library system through an exploration of the reading devices that adolescents choose, how they utilize technology to read, and how they use public e-resources [8]. Vahide Yiğit Gençten and Filiz Aydemir [2023] state that an increasing amount of study is exploring how technology tools might effectively engage young children in reading activities due to the growing integration of technology into early years schooling. They highlighted how reading activities with technology assist children's development of various abilities and explain the advantages that result from them. The activities can strengthen language abilities by expanding resource access [9].

Ali Mufron, Kailie Maharjan, Eladdadi Mark, and Embrechts Xavier [2024] discuss how technology integration can be effectively used in the classroom and discuss the impact of technology integration in learning on increasing student engagement. The direct effects of technology integration on learning will be felt by students in terms of their use of it and how it affects their activities, creativity, and goals [10].

According to Americo N. Amorin, Lieny Jeon, Yolanda Abel, Eduardo F. Felisberto, Leopoldo N. F. Barbosa, and Natalia Martins Dias [2020], touchscreen interfaces on mobile devices like smartphones and tablets make them more user-friendly and intuitive. Using gamification techniques like the ones used in this intervention could boost student motivation and learning at a large scale compared to traditional computer-based laboratories; mobile devices like tablets and smartphones are far less expensive and require less upkeep [11].

Saeed S. Alqahtani [2020] stated that the literature review looks at technology-based interventions for children struggling with reading. Computer programs and other technology-based interventions can assist struggling readers in improving their fluency. However, less focus was placed on tablet-based interventions and vocabulary skills. These results are consistent with your study's emphasis on using technology to advance reading literacy. Teachers can design engaging learning experiences that enhance reading development using technology tools. Technology can empower young readers and close literacy gaps by utilizing interactive apps, e-books, or adaptive platforms. Take adaptive e-books, for example, which change the difficulty level according to a child's development. These tailored encounters with interactive components and fast feedback can scaffold learning [12].

Min Park Mize and Melissa Yujeong Martin [2023], the study showed that using technology in fluency interventions can benefit struggling readers, indicating that technology is effective in enhancing reading fluency. Teachers should use a rubric when using tablets due to the lack of instructional components. They recommend further research to investigate how effectively technology-assisted interventions improve reading fluency. Some studies analyze the impact of general reading interventions, but additional research is necessary regarding fluency and technology-based fluency interventions. With technology constantly evolving, more studies are required to investigate using tablets and apps to improve reading fluency [13].

Marie Palmer [2021] studied how public libraries are changing in today's world. The article talks about how libraries are moving away from just places for books and becoming community centers and areas for learning. It was further stated that libraries need to change to fit the needs of their communities by using technology, working with other groups, and offering new and exciting services. Libraries can remain important for education, helping people get involved in their communities and bringing people together [14].

The study by Mustafa Altun1 & Hassan Khurshid Ahmad [2021] mentioned that technology is crucial in advancing the teaching and learning methods in educational institutions, particularly in English language education. It can help the teacher deliver a more effective lesson to students. Technology plays a vital role in education, enabling teachers to incorporate multimedia elements like English videos, songs, movies, and theatrical shows [15].

Comfort Makwanya and Olabanji Oni [2019], the findings show observational evidence that electronic books, other programming books, and online instructional tools give more students options. This stated that e-books allow users to access the specific articles desired through ejournals. Electronic books are becoming the choice for students because they are convenient, portable, quickly searched, less weighed, and provide remote access [16].

Manju Kanwar Rathore and Reeta Sonawat [2020], the study investigates how students' digital literacy in academic libraries is affected by the integration of technology. It was discovered that integrating technology into the classroom can significantly enhance students' proficiency with technology-based learning, database utilization, and online information searching [17].

Xue Wen, Shauna M. Walters's [2022] study thoroughly examines how technology can enhance reading skills, showing that technology interventions greatly benefit reading comprehension, vocabulary, and fluency. With the evolving literacy landscape and the significance of early writing skills, incorporating technology and writing in elementary schools offers a promising solution to address this issue [18].

In the study of Tammy S. Gruer and Karen M. Perry [2020]. Incorporating technology in reading has dramatically changed how individuals interact with text and improved their literacy abilities. They emphasize that reading has become more accessible and enjoyable, especially for young readers and those with reading challenges, due to technological advancements like e-books, audiobooks, and online databases. The research also highlights how technology expands access to reading materials, promotes inclusivity, and can support the creation of a more inclusive and literate society by being incorporated into public libraries. Furthermore, multiple library initiatives focused on improving digital literacy have effectively promoted reading using technology [19].

Mirza Quratulain, Pathan Habibullah, Khatoon Sahib, and Hassan Ahdi [2021] examine how digital technology and social media affect the reading habits of students at Mehran University of Engineering and Technology in Pakistan., influenced by the ease and availability of digital platforms. Although social networking sites may divert students from their academic reading, they also enable the exchange and debate of educational material. The research emphasizes a rise in how often people read because of digital availability, yet it recognizes a drop in the thoroughness and excellence of reading. Barriers to focused reading, such as digital distractions and information overload, were identified as challenges. The results indicate that although digital technology makes reading more accessible, measures are required to enhance reading quality and minimize distractions [20].

Effectiveness of Technology Integration in Promoting Reading Literacy

According to Brayan Diaz, Nussbaum, Miguel, Greiff, Samuel, and Santana, Macarena. [2024], the precise design of the technology and the pedagogical strategy used to determine how effective it is to teach reading comprehension. Well-designed digital books perform better than reading on paper. Supporting reading habits, student-centered technology-based activities are linked to improved reading performance. Proactive policy actions ought to assist educators and schools in implementing student-centered technology integration strategies [21].

The study by Pálsdóttir [2019] concludes that printed and electronic study materials have advantages and disadvantages. The best format for students may depend on their preferences and learning styles. It suggests that technology can be a valuable tool for learning and that students should be encouraged to explore printed and electronic study materials to find the best format for their needs [22].

According to Maria Cahill, Erin Ingram, and Soohyung Joo [2023], using technology is common among children and families, leading parents and caregivers to worry about the safety and development of children when using technology. If children's librarians are to confidently assume the media mentor role and have the necessary knowledge and skills, more focus and support on professional development and resources are essential for guiding technology use in children and families [23].

The study by Muhammad Rafi, Zheng JianMing, and Khurshid Ahmad [2019] stated that library technological integration greatly enhanced students' proficiency in technology-based learning. The academic libraries have incorporated digital tools to improve students' digital literacy and technical abilities. It underscores the significance of training and instructors' crucial role in preparing students with the necessary skills to utilize digital resources effectively. This integration improves students' skills in finding information online and using database resources efficiently [24].

The research study by Pagore Ranjeet [2022] explores ways mobile technology can improve library services. This likely includes mobile apps for searching library catalogs, accessing digital resources, and enhancing user engagement. The potential of mobile technology is to increase access to e-books and digital resources and improve user experience through convenient mobile apps. This demonstrates awareness of the broader discussion on integrating technology in libraries, specifically through mobile applications, and strengthens the foundation for your research on promoting reading literacy through technology at Iriga City Public Library [25].

Millie Nne Horsfall and Augustine Chineme Opurum [2023] state that emerging intelligent technologies for brilliant school libraries are covered in the article. Emerging intelligent technologies are advancements in hardware, software, or applications that allow librarians to effectively prepare and present information content to users or learners in an engaging manner, therefore facilitating the delivery of information services. It depicted an intelligent school library as an interactive hub for learning and other educational pursuits within the school library. Among the growing intelligent technologies that can be employed for brilliant school libraries are speed reading software, Plotagon, and math problem-solving software [26].

Oskar Hernandez Perez, Oskar Fernando Vilarino, and Miquel Domenech [2020] suggest creating a Library Living Lab, where librarians can try new things like technology, services, and ways to connect with the community—the importance of knowing what the community needs. By customizing library services to fit these needs, libraries can better serve everyone. They can make the library experience better through technology. Things like digital literacy programs, maker spaces, and places for people to work build community together. These technologies give people power and encourage lifelong learning. But it also points out opportunities for growth and change. Libraries must be flexible and ready to adapt to people's wants [27].

The study of Khurram Shahzad and Shakeel Ahmad Khan [2023] aimed to identify new elearning technologies and the difficulties they face in implementing them. It has provided suggestions and workable solutions for successfully integrating cutting-edge technology in elearning, impacting society. It is recommended that managerial implications be supplied for creating effective policies for e-learning technology implementation, which will support the development of creative, sustainable competence in library staff and the deployment of intelligent library services [28].

The study of Charisse Krsyel C. Contago, Charry A. Lanas, Kim M. Travero, and Lloyd Matthew C. Derasin [2024] states that education is on the verge of a revolution, marked by the quick advancement of technology and the growing ubiquity of digital media. A potent and captivating teaching tool has surfaced as a possible enhancer of pupils' reading comprehension abilities. The educators and researchers grapple with the challenge of cultivating proficient readers to investigate the multifaceted role that digital storytelling plays in developing reading comprehension among those age groups. Reading comprehension is a fundamental skill that underpins academic success and lifelong learning [29].

The research conducted by Alpaslan Toker and Saidou Aminou [2019] explores the reading behaviors of university students; it points out various obstacles to regular reading routines, including language difficulties, the negative influence of the internet, disinterest, dependence on television, and limited book availability. The crucial significance of reading for personal and academic growth, emphasizing its role in acquiring language, social connections, happiness, and academic achievement communication, and sharing information and ideas. They propose strategies to enhance reading behaviors, such as expanding book availability, fostering a reading culture, and combating the adverse effects of digital diversions [30].

The study of Sultan Sarwat, Hussain Irshad, and Fatima Shabbih [2020] investigates how internet usage relates to social connectedness, life satisfaction, and academic success in college students. It recognizes that overusing the internet can harm these aspects, but using it in moderation can improve them. It was found that the significance of using the internet in a balanced way is to preserve social connections, happiness, and academic achievement. It stresses that although the internet is helpful for information and communication, too much usage can result in isolation and reduced satisfaction with life. Suggestions include advocating for moderate internet usage among students to improve their social connections, happiness, and academic success. They also urge educational institutions to offer advice on healthy online behavior and incorporate internet usage into academic tasks to boost students' overall health [31].

The research by Lefose Makgahlela and Amogelang Molaudzi [2020] points out an ongoing absence of a reading culture among South African students, especially in higher education institutions. It highlights the significance of forming reading habits that last a lifetime, beginning in the home and extending into school and beyond. It suggests that the University of Limpopo Library should add more leisure reading materials to promote reading for enjoyment and relaxation among students [32].

The Research conducted by Ahmad Shabir, Mr. Dar Ahmad Bilal, and Mr. Lone Ahmad Javed [2019] analyzed studies of the reading behaviors and opinions of college students at two institutions in Anantnag, Jammu and Kashmir, with a particular emphasis on gender variations. It also emphasizes how social media and digital spaces have affected students' reading routines, decreasing traditional reading habits [33].

Lidia Acuna Torres, Rocio Flores Pezo, Blanca Ines Laso Garcia, and Milko Raul Rivera Campano [2024] analyzed the positive impact of technological tools on students in improving reading comprehension and developing digital skills in both educators and students. This would ensure meaningful, high-quality learning, with technological tools as learning mediators, contributing to reducing the digital divide [34].

Alana Oif Telesman, Moira Konrad, Gwendolyn Cartledge, Ralph Gardner, and Morris Council [2019] claimed that the level of reading success a student achieves has significant consequences for their future, and it is crucial to prioritize addressing any reading issues early and thoroughly. Hence, early intervention programs aim to improve students' reading comprehension scores and boost their skills to face obstacles in the school setting [35].

According to Ronen Kasperski, Michal Shany, Tal Erez Hod, and Tami Katzir [2019], personalized computerized reading programs can positively impact children's reading skills and self-esteem. These programs tailor instruction to each student's needs, potentially leading to better reading outcomes and a more substantial confidence in reading ability. It highlighted the role of technology in enhancing children's academic performance and self-awareness [36].

Natalia Kucirkova [2019] declared that technology is associated with student motivation in both educational and academic fields. Its online learning platform has boosted children's drive and enthusiasm for learning. Digital books offer a suitable platform to enhance learners' motivation through their interactions with the text. Likewise, interactive elements in books help children grasp the main storylines and the connection between the story and additional activities [37].

According to Kirsten W.J Touw, Bart Vogelaar, Merel Bakker, and Wilma C.M. Resing [2019], teachers frequently employ interactive boards, video support, tablets, and mobile technology to enhance their teaching. These tools create engaging learning environments that stimulate students' motivation and interest in reading. By addressing individual learning needs, these technologies assist students in overcoming diverse reading challenges. Moreover, focusing on technology fosters a student-centered system, encouraging problem-solving and critical thinking, while personalized computer instruction further motivates students and strengthens their commitment to surpassing academic challenges. [38].

Arumugam Raman and Raamani Thannimalai [2019] state that with technology advancing quickly, artificial intelligence and the Internet of Things transform school leadership roles, teaching methods, and classroom designs. School administrators must motivate and assist educators in incorporating technology technology to enhance their teaching. And making it into their teaching, particularly as the Internet of Things School leaders must stay updated literacy activities to increase interest in reading. on new technologies due to the everyday use of intelligent whiteboards and other interactive digital media in interactive learning environments. [39].

Nurman Samehuni Gea, Upaya Bakti Murni Gulo, and Arozatulo Bawamenewi [2024] state that reading is vital for developing insight, knowledge, and critical thinking skills. Therefore, schools should organize literacy activities to increase interest in reading. Digital literacy is understanding and using information from various digital sources, technology, and digital devices in different situations [40].

This article discusses how technology has made education more personalized and efficient. Digital resources can be beneficial for children's literacy. Some programs can support children with phonics, fluency, and vocabulary. Technology can also support equity by providing resources to children who may not have them at home. Technology allows parents and educators to tailor learning experiences to meet individual needs.

Synthesis of the State of the Art

The study of Yu-Zhou [2021], Muntaha Ali Mohammad Al Momani [2021], Wen-Chi [2024], Lopez-Escribano [2021], Smith [2020], Noordan [2022], Goek [2023] Loh [2022] Gençten [2023] Mufron [2024] were related to the study stated that technologies can enhance users reading ability and provide a complete resource to users who can access libraries technology resources.

Amorim [2020], Alqahtani [2020], Mize [2023], Palmer [2021], and Altun1 [2021] were related to a study that explained how technologies could enhance the reading literacy of a user. The survey of Makwanya [2019], Rathore [2020], Wen [2022], Gruer [2020], and Mirza Quratulain [2021] explained how digital literacy in academic libraries is affected by the integration of technology and the overview of the effectiveness of technology in improving reading literacy.

Diaz [2024], Pálsdóttir [2019], Cahill [2023], Rafi [2019], Pagore [2022], Horsfall [2023], Pérez [2020], Shahzad [2023] Schmitt [2019] Toker (2019) was related to the study that was determined through technologies users can enhance their reading ability because of different reading features of each reading applications. This is how effectively technologies are used for reading comprehension.

According to authors Sarwat [2020], Makgahlela [2020], Shabir [2019] Acuña-Torres, [2024] Telesman, [2019] were related to the study that explained the impact of technological tools on the students in improving reading comprehension and the reading achievement that has a significant implication for students' future success.

In addition to the study of Kasperski [2019], Kucirkova [2019], Touw [2019], Raman [2019] Samehuni Gea [2024] Claimed that technology is connected to students' drive to learn and academic success by providing tools to establish captivating educational settings that enhance students' interest and motivation in reading.

Gap Bridge of the Study

The fundamental object of this study is to integrate technology in promoting reading literacy in the Iriga City Public Library. As researchers, we will analyze and evaluate the related literature and studies from different journal articles to identify and observe the possible gap in the study.

1. There is no current study of integrating library technologies conducted in the Philippine libraries, mainly focusing on its challenges.
2. There are no local studies conducted yet in Iriga City Public Library in Camarines Sur regarding technology integration in promoting reading literacy.
3. no existing study will identify the challenges encountered in technology integration in the Iriga City Public Library.

Due to these gaps, researchers aimed to study technology integration to promote reading literacy at the Iriga City Public Library.

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Chapter 3

RESEARCH METHODOLOGY

This chapter overviews the research methods and procedures applied in this study. The technique or choice of respondents to be used for the analysis and interpretation of responses from participants in this field is also covered, as well as data collection instruments and documentary analyses that will be used during the study.

Research Design

The researchers used quantitative data to complete an overview of the current technological integration promoting reading literacy in the Iriga City Public Library. This researcher used the descriptive method to accurately and methodically define a situation, population, or phenomenon. This method is crucial to identifying all the needed data by Pritha Bhandari [2023]. Descriptive method design aims to employ a purposive sampling technique to select participants who visit the library and have experienced the technology integration initiatives. This study improves library services, ensures equitable access to technology, and supports community users learning and development in the digital age [1].

Theoretical/Conceptual Framework

The Technology Acceptance Model (TAM) developed by Davis [1989] is referenced in Robert Allen's research and aims to 'offer better explanations about technology integration' of the Iriga City Public Library in promoting reading literacy. In this study, perceived usefulness, as defined by TAM, is the intention of integrating technologies and how their usage is perceived [2]. The belief is that with the aid of technology resources, it would be easier for the Iriga City Public Library patrons to access and utilize library services. This study will explore how perceived usefulness, as influenced by TAM, affects the patrons' adoption to the technology integration and the use of technology-based library resources.

Operational Framework

Based on this theory, the *Technology Acceptance Model (TAM)* is a comprehensive framework that explains how people use technology. TAM aims to predict and explain user behavior towards technology adoption, concentrating on two main elements, perceived usefulness and perceived ease of use, in incorporating technologies into the Iriga City public library to enhance reading skills. the profile of library users, strategies for technology integration, challenges faced during implementation, and the specific context of the Iriga City public library.

Figure 1. The technology integration and the Iriga City Public Library are the independent variables. In contrast, the dependent variables were the profile of the respondents, strategies, and challenges.

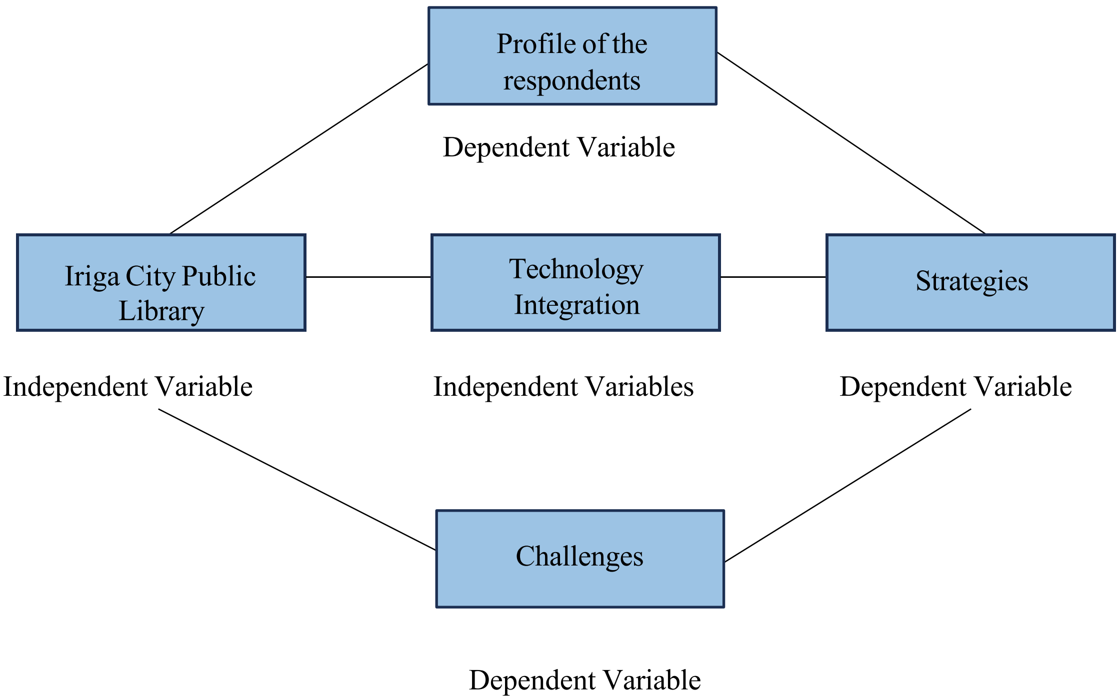

Figure 1: **<u>Operational Framework of the study</u>**

The Respondents

The respondents of this study were individuals within the community, as well as different and diverse users of Rizal St, Iriga City Camarines Sur. Iriga City Public Library

The researchers used Slovin's formula and an Ideal Margin of error/significant level of 5%, commonly employed when selecting a respective sample from a large population, By calculating the appropriate sample size of 353, population of 3,000 a week and margin of error of 5% or .05.

The users were chosen through Random sampling from the total population of Iriga City Public Library within a month: approximately 150 users per day or 3000 users in the library monthly.

Data Gathering Tools and Procedures

To efficiently gather the required data of the population of Iriga City Public Library, the researchers used a survey questionnaire checklist as it was the most suitable data collection tool to gather the necessary data for the successful conclusion of this research.

To prepare the questionnaire, the researcher gathered and reviewed related literature and studies to construct the questions. The questionnaire is composed of Three (3) parts. Part 1 will identify the respondents' profiles regarding age, gender, and type of library user. Part 2 determined the technologies that can be integrated into the Iriga City Public Library.

Before being distributed, the developed checklist questionnaire was reviewed by the panel, adviser, instructor, and consultant. The questionnaire was reviewed, and a practice run was performed. It was handed to 20 random users from the Iriga City Public Library.

The approved survey was distributed to all 355 respondents. The researcher requested the respondent's approval before answering the survey questionnaire. The questionnaire was retrieved after the respondents filled it out and sent it to the researcher. The questionnaire was extensively reviewed, tabulated, and identified for analysis and interpretation. The data collection was also kept confidential.

To assist in the analysis process, the researchers gave the survey questionnaire results to the statistician to compute all the data collected. The next day, the statistician informs the researcher that he’s done computing the results, utilizing appropriate treatment. This partnership guarantees the precision and thoroughness of the data analysis procedure.

Statistical Treatment of Data

After collecting the responses from the questionnaire, the researchers processed and analyzed the data to derive meaningful results. Following statistical techniques were employed to analyze the data.

Slovin’s formula

Slovin’s formula is used to determine the minimum sample size for a study. It helps decide how many respondents to include to ensure the research findings are reliable. The researchers can be confident that their results are accurately represented in the research findings.

Formula:

$$n = \frac{N}{1 + N\,e^2}$$

Where:

n - Sample size

e - Desired margin of error

N - Represents total population

Percentage Technique

The technique was employed to verify the calculated percentage equivalent of the collected responses. The researchers used the formula to determine the percentage of respondents who selected each answer choice for a given answer. The percentage method, a widely used statistical representation technique, employed to visualize the results.

Formula:

$$P = \frac{f}{n} \times 100$$

Where:

P - Percentage

f - Frequency

n - Number of respondents

100 – constant

Weighted Mean

It was used to acquire the findings and provide an object interpretation, such as the technologies that can be integrated into Iriga City Public Library and the challenges Iriga City Public Library encountered in technology integration in promoting reading literacy along the indicators.

Formula:

$$WM = \sum \frac{\sum fx}{N}$$

Where:

WM - represents the weighted mean or the result of the equation

$\sum$ - Summation

x - scale value

N - Sum of frequencies of number of respondents

Likert Scale

This scale assesses opinions and determines the respondent's level of agreement or preference towards a specific statement in their research study. Alternatively, the researcher utilized a four-point scale consisting of a scale-weighted mean for calculating respondent averages, verbal interpretation, and description.

Four Points Likert scale to assess the technologies that can be integrated with promoting reading literacy in the Iriga City Public Library and to determine and summarize the result of the challenges Iriga City Public Library encountered in technology integration promoting reading literacy.

For the best technology to integrate in the Iriga City Public Library, the researchers used the following scale.

Scale	**Mean Range**	**Verbal Interpretation**
4	3.26 – 4.00	Strongly Agree
3	2.51 – 3.25	Agree
2	1.76 – 2.50	Disagree
1	1.00 – 1.75	Strongly Disagree

Notes

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Chapter 4

RESULTS AND DISCUSSIONS

In this chapter, the data collected in the study on Technology integration in promoting reading literacy in Iriga City Public Library was presented, analyzed, and interpreted. These technologies can be integrated into the Iriga City Public Library and challenges the Iriga City Public Library encountered in technology integration in promoting reading literacy. To ensure clarity, the information is displayed in a table format.

1. Respondent's Demographic Profile

This part discusses the profile of the respondents in terms of age, gender, and types of library users, including the distribution frequency and percentage of the data, including the total number of respondents.

Table 1
RESPONDENTS PROFILE

Pr file		FREQUENCY	PERCENTAGE
Age	10-20	90	25.50
	21-30	**165**	**46.74**
	31-40	40	11.33
	41-50	36	10.20
	51 Above	22	6.23
Gender	Male	134	37.96
	Female	**219**	**62.03**
Type of Library User's	**Student**	**239**	**67.70**
	Professional	21	5.94
	Parent	44	12.46
	Others	49	13.88
Total Number of Respondents = 353			

Table 1 shows the frequency distribution and percentage of users' profiles of Iriga City Public Library in terms of age corresponding age. Out of 353 respondents, 90 or 25.50%, ages from 10 to 20 years old, 165 or 46.74%, ages from 21 to 30 years old, 40 or 11.33%, ages 31 to 40 years old, 36 or 10.20%, ages 41 to 50 years old, moreover there are only 22 respondents or 6.23% that are 51 years old and above in the total of 100%. The result shows that among the indicators, 21 to 30 years old, primarily among them are students who are the primary users of the Iriga City Public Library. It determines that the library resources comprise students doing each other's research, assignments, and activities.

Table 1 shows the frequency distribution and percentage of users' profiles of Iriga City Public Library in terms of gender with the corresponding percentage. Of 353 respondents, 134, or 37.96%, are male, while 219, or 62.03%, are female in 100%. The results revealed that among males and females, primarily female users are the primary users of the Iriga City Public Library.

Parishmita Hazarika [2021] stated that it was discovered that libraries offer women critical environments for solitude, education, and social interaction, especially for those who do not have access to such spaces in their everyday routines. This results in women visiting libraries more often than men. Libraries provide a secure, peaceful space for women to explore their passions, utilize learning materials, and interact with society. It emphasizes how libraries enhance knowledge and skills, support individual growth, and build community. Libraries are popular among female users because they offer opportunities for reading, studying, attending workshops, and participating in events. Overall, the article highlights the significance of libraries in aiding women's educational and personal development [1].

The data in Table 1 displays the breakdown and percentage of different types of library users at Iriga City Public Library, including students, professionals, parents, and scholars/researchers. Among 353 participants, 67.70% are students, 5.94% are professionals, 12.46% are parents, and 13.88% are classified as "Others" in the overall percentage of 100%. The findings showed that most library patrons are students, accounting for 239 individuals or 67%.

The study by Maria Juanita R. Macapagal [2023] examines how public libraries in the Philippines contribute to meeting students' educational requirements. It emphasizes the importance of libraries providing essential resources such as books, digital materials, and study spaces. In addition, the study emphasizes the significance of libraries in encouraging a culture of reading and lifelong learning in students. However, it suggests various improvements, such as increasing library funding, updating facilities, and enhancing digital resources to serve students' needs better. This extensive assessment offers a comprehensive look at the strengths and areas for growth within the Philippine public library system [2].

2. **Technologies can be integrated to promote reading literacy in the Iriga City Public Library.**

This part discusses the technologies that can be integrated into the Iriga City Public Library, which explains the indicators, including the weighted mean, verbal interpretation, ranking, and the average weighted mean.

Table 2 **TECHNOLOGIES THAT CAN BE INTEGRATED INTO IRIGA CITY PUBLIC LIBRARY**

Indicators	Weighted mean	Verbal Interpretation	Ranking
Electronic books	3.98	Strongly Agree	2
Audio-books	3.96	Strongly Agree	5
Electronic Gadgets	**3.99**	Strongly Agree	**1**
Educational E-games	3.97	Strongly Agree	4
Reading applications	3.96	Strongly Agree	5
Interactive Displays	3.92	Strongly Agree	7
Online Tutoring Platforms	3.91	Strongly Agree	8
ICT Facilities	3.98	Strongly Agree	2
Average Weighted Mean	**3.96**	Strongly Agree	

Table 2 shows the overall average weighted mean of 3.96, verbally interpreted as strongly agreeing. The Iriga City Public Library user community is optimistic about integrating technologies to enhance library services and programs. This integration aims to improve access to diverse information, foster reading literacy, and promote an inclusive learning environment.

The technologies that can be integrated into Iriga City Public Library are ranked based on their average weighted mean. Rank 1 with a weighted mean of 3.99 is electronic gadgets with the verbal interpretation of Strongly agree, rank 2 with a weighted mean of 3.98 is ICT facilities and electronic books with a verbal interpretation of Strongly agree, rank 4 with a weighted mean of 3.97 is electronic E-games with the verbal interpretation of Strongly agree, rank 5 with a weighted mean of 3.96 is reading applications and audiobooks with verbal interpretation of Strongly agree, rank 7 with a weighted mean of 3.92 is interactive displays with verbal interpretation of Strongly agree. Rank 8, with a weighted mean of 3.91, this is an online tutoring platform with verbal interpretation of strongly agree. This means that utilizing electronic gadgets such as smartphones, tablets, and computers in the daily operation of the library and These electronic devices provide unparalleled accessibility to possible library educational materials, allowing users to engage with content anytime and anywhere.

These findings were related to the study of Nataliia V. Marchenko, Halyna I. Yuzkiv, Ivanenko, Iryna M. Ivanenko, Olena M. Khomova, and Kateryna M. Yanchytska [2021], which is emphasizes the technology integration promoting reading literacy to discover electronic gadgets. Users can download educational application tools to improve their spelling, reading, pronunciation, and vocabulary, practice making sentences, and find more interactive features [3].

3. Challenges Encountered by Iriga City Public Library in Technology Integration Promoting Reading Literacy

This part discusses the challenges Iriga City Public Library encountered in technology integration promoting reading literacy, and to which explains the indicators, weighted mean, verbal interpretation, ranking, and average weighted mean.

Table 3
CHALLENGES ENCOUNTERED OF IRIGA CITY PUBLIC LIBRARY IN TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY ALONG THE INDICATORS

Indicators	Weighted mean	Verbal Interpretation	Ranking
Limited ICT Skills	3.74	Strongly Agree	8
Limited Fund for Technology Integration	3.90	Strongly Agree	2
Library Space and Capacity	3.80	Strongly Agree	6
Internet Connectivity	3.86	Strongly Agree	3
Usage Policies	3.79	Strongly Agree	7
Lack of IT Staff	3.81	Strongly Agree	5
Technology Safety	3.83	Strongly Agree	4
Technology Accessibility	**3.92**	**Strongly Agree**	**1**
Average Weighted Mean	**3.83**	**Strongly Agree**	

Table 3 shows that the average weighted mean was 3.83, with a verbal interpretation of strongly agreeing. This means that Iriga City Public Library users recognize the challenges of integrating technologies into library services. This suggests that while there is an intense desire for technological innovation, users also may experience difficulties in this transition. The challenges encountered by Iriga City Public Library in technology integration promoting reading literacy, rank 1 with a accessibility with verbal interpretation of Strongly agree, rank 2 limited fund for interpretation of strongly agree, rank 3 with a weighted mean of 3.86 is Internet connectivity with verbal interpretation of strongly agree, rank 4 with a weighted mean of 3.83 is technology safety with verbal interpretation of Strongly agree. Rank 5 with a weighted mean of 3.8, 1 is lack of IOT staff with verbal interpretation of strongly agree, rank 6 with a weighted mean of 3.80 is library space and capacity with verbal interpretation of strongly agree, rank 7 with a weighted mean of 3.79 is usage policies with verbal interpretation of Strongly agree, rank 8 with a weighted mean of

3.74 is Limited ICT Skill with verbal interpretation of Strongly agree. This means the Iriga City Public Library users have identified technology accessibility as the most critical issue to address when integrating technology into library services. Most users may find it challenging to access and use the various technologies inside the public library since some of them are unfamiliar or unsure of how to use and navigate correctly, which sometimes may lead to a fear of damaging the technology.

The study of Gurumurthy Padmamma S. Kuri [2023] discusses the impact of information and communication technology (ICT) on libraries regarding their accessibility and functionality. ICT has increased the accessibility of library resources by digitizing them and creating userfriendly interfaces, which benefit students, remote users, and people with disabilities by ensuring they have equal access to information. Ensuring easy user access aims to improve their overall experience and cater to their information requirements. At the same time, ICT enhances the efficiency and effectiveness of library services through advanced search systems, automated cataloging, and resource management tools, making it more straightforward for users to access and utilize resources. Using technology correctly increases user satisfaction by providing a smooth and practical library experience [4].

Notes

[1] Parishmita Hazarika. 2021. An assessment of reading habits and use of public library resources by rural women of sivasagar district: A survey. *Library philosophy and practice (e-journal).* (December 2021). https://digitalcommons.unl.edu/libphilprac/6884

[2] Maria Juanita R. Macapagal. 2023. Status report of Philippine public libraries and librarianship. *National Library of the Philippines.* Retrieved from http://web.nlp.gov.ph/nlp/sites/default/files/14Jun2019/StatusofPhilippinePublicLibrariesandlibrarianship.pdf

[3] Nataliia V. Marchenko, Halyna I. Yuzkiv, Iryna M. Ivanenko, Olena M. Khomova, and Kateryna M. Yanchytska. 2021. Electronic resources for teaching ukrainian as a second language. *International Journal of Higher Education,* 10(3), (February 2021) 234-245. DOI: https://doi.org/10.5430/ijhe.v10n3p234

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Chapter 5

SUMMARY FINDING, CONCLUSION AND RECOMMENDATION

This chapter discussed and presented the summary of the study's findings, conclusions, and recommendations. The conclusions and recommendations are based on the findings obtained by the researchers.

Summary

This study aimed to determine the integration of technology in promoting reading literacy at the Iriga City Public Library. The findings of this study will benefit the public library and its organization. This study examines the technologies that can be integrated or available at the Iriga City Public Library. We also investigate the potential challenges encountered by the Iriga City Public Library in technology integration to promote reading literacy among library users.

Findings

After gathering the needed information and data, we, the researchers, distributed a survey questionnaire; the following are the study's key findings:

1. Quantitative results revealed that 165 out of 355, or 46.74 %, were in the age bracket 21-30, the majority of the respondents. Most respondents were female, with 219 respondents, or 62.03%, and 134 respondents, or 37.96%, were male. The results show that most respondents were students, with 239 or 67.70%.
2. The technology recommended for integration in Iriga City Public Library is ranked according to their average weighted mean. The ranked technology, with a weighted mean of 3.99, is electronic gadgets, indicating substantial agreement among respondents on their importance and following with the ICT Facilities and electronic books, both have a weighted mean of 3.98, also reflecting strong agreement. Electronic e-games ranked 4th with a weighted mean of 3.97, while reading applications and audiobooks ranked 5th with 3.96. interactive display with a 3.92,

3. and online tutoring platforms with a weighted mean of 3.91. these emphasize the significant role of electronic gadgets, such as smartphones, tablets, and computers, in enhancing library operations and improving users' reading abilities. These devices offer accessibility to educational materials, enabling users to engage with content anytime and anywhere, thus supporting their learning and literacy development.
4. Iriga City Public Library users know the challenges in integrating technologies into library services. Technology accessibility emerged as the most critical issue, with a weighted mean of 3.92, suggesting that users find it chathe various technologies available in the library. This could be due to unfamiliarity or uncertainty, sometimesfunds for technology integration, with a weighted mean of 3.90, safety with 3.83. additionally, the lack of IT Staff 3.81, limited library space and capacity 3.80, usage policies 3.79, and limited ICT skills among staff 3.74 are identified as the barriers to effective technology integration. Addressing these challenges is essential for successfully promoting reading literacy within the library, ensuring that all users can benefit from the available technological resources.
5. The researcher proposed a contingency plan as the output of this study, which explains and plan how the challenges can have a proper solution for technology integration promoting reading literacy in the Iriga City public library.

Conclusions

Based on the findings, the following conclusions were drawn:

1. The age group of 21–30 received the highest ranking, and most respondents were female students' users of the Iriga City Public Library.
2. The highest-ranking indicator in determining what technologies can be integrated in Iriga City Public Library was the "Electronic Gadgets," which concludes that the Iriga City Public Library

users have a strong preference for mobile devices and other portable technology to integrate for improving the library services for reading literacy and to accommodate the needs of library patrons in the community.

3. The study identified that technology accessibility as the highest rank issue in integrating technology to promote reading literacy at Iriga City Public Library. Although users see the potential benefits of new technologies, they may also feel anxious and hesitant about using them. The researchers concluded that with adequate user engagement and orientation, library patrons can effectively use these technologies. Proper guidance and support are crucial to ensure that these advancements are embraced and maximized, ultimately enhancing the reading literacy experience for all users.
4. The Iriga City Public Library's contingency plan integrates secure digital tools to enhance information access and reading literacy skills and foster lifelong learning for public library users. The contingency plan aims to utilize technology in a safe and beneficial setting, which aligns with the library's commitment to providing excellent services and promoting community growth.

Recommendations

Based on the findings and conclusions, the following recommendations are derived:

1. Establishment of ICT section/ media center: Establishment of an ICT Section or Cyber Room. This section should have a dedicated space with computers, tablets, and high-speed internet access. This offers a variety of digital resources, including electronic books, audiobooks, online databases, and educational software for reading. The specialized facility is designed to be accessible to all library patrons and is equipped with technologies that can enhance the reading skills of library users. This center should strategically locate near the main entrance or lobby of the Iriga City Public Library for easy accessibility. This section offers digital resources such as computer units and touchscreen tablets supporting various reading and learning activities.

This specialized facility will help enhance users' reading capabilities.

2. Established a comprehensive evaluation process to select and test high efficiency and specifications for the public library patrons and staff, prioritizing reputation, durability, performance, and user-friendliness. Ensuring the procured electronic gadgets are compatible with existing systems and support diverse digital reading tools. The committee must conduct thorough market research, gather user feedback, and choose quality devices—partner with reputable suppliers for the best deals and warranties. Test gadgets through a pilot program before full-scale procurement through this approach will improve future readers' reading literacy and user experience.
3. Upgrade the Wi-Fi infrastructure to support more users and establish designated zones with high internet bandwidth. Implement a Wi-Fi management system to ensure fair access, promote online reading activities, and encourage the use of digital reading materials. This measure will enhance internet connectivity of all devices inside the library and improve the overall user experience of future readers.
4. Hire IT personnel who are proficient in hardware and software digital technologies. Their expertise ensures library users have the support to access and utilize digital reading resources effectively. Conduct a training session for all library's IT staff that will enable them to manage and navigate the digital reading resources and infrastructure, which involves maintaining systems of various reading tools, troubleshooting issues, and continuously enhancing the digital reading infrastructure, ensuring the library's resources are current, fully functional, and supportive of users' reading development.
5. Provide a hands-on workshop on effectively using reading tools inside the library. These will cover various topics, such as providing the library users an overview of its digital resources and function, how to access their library portals, online platforms, and reading applications,

and the primary navigation of various reading tools. By making these programs accessible to everyone, library users can significantly improve their reading skills, gain confidence in utilizing digital reading resources, and fully benefit from the library offerings to support their reading and literacy development.

6. The proposed recommended output is a contingency plan entitled “Contingency Plan for Promoting Reading Literacy by technology integration in Iriga City Public Library” that develops comprehensive strategies to manage and mitigate technological disruptions, enhances community reading skills through technology integration, addresses the challenges of technology adoption, and promotes reading literacy by implementing technologies in the Iriga City Public Library. This plan specifically focuses on improving reading literacy in the community. It outlines the challenges, proposed actions, responsible personnel, expected responses, and the target dates for implementation. This structured approach addresses the challenges of technology adoption for reading literacy in Iriga City Public Library, ensuring that each step is taken to achieve the desired outcome for enhancing reading literacy among library patrons.

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Talesin L. Smith and Emily B. Moore. 2020. storytelling to sensemaking; a systematic framework for designing auditory description display for interactives. *In proceedings of the 2020 chi conference on human factors in computing systems (CHI '20).* April 23, 2020, Honolulu, HI, USA. Association for Computing Machinery, New York, NY, USA, 1-12. DOI: https://doi.org/10.1145/3313831.3376460"0

Wen-Chi Hu and Shih-Tsung Hsu. 2024. The future of EFL reading: How digital technology is changing the game. *In proceedings of the 2023 7th international conference on computer science and artificial intelligence (CSAI'23)*. March 14, 2021, Beijing, China. Association for Computing Machinery, New York, NY, USA, 339-345. DOI: https://doi.org/10.1145/3638584.3638668

APPENDICES

Appendix A
APPROVED LETTER

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED

COLLEGE of
COMPUTER STUDIES

May 21, 2024

DR. AMADO A. OLIVA, JR.
College President, Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur

Dear Sir,

Greetings of Peace!

We, the undersigned 4th year students are presently conducting research entitled "TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY" in partial fulfillment of the requirements for the degree of Bachelor of Library and Information Science at Camarines Sur Polytechnic Colleges.

In the connection of the above, may we respectfully request permission from your good office to allow us to conduct a survey through questionnaire in Iriga City Public Library users for the purpose of gathering information about the technology integration in promoting reading literacy in Iriga City Public Library. Your approval will be significance of the success of our study which will provide the needed information and response for our research. Rest assured that all information to be obtained will be treated with utmost confidentiality.

Your response and time are greatly appreciated. We are hoping that this request will merit your favorable approval.

Thank you very much and God bless!

Sincerely yours,

JHOEMIE A. YAON

BEVERLY G. JACOB

MA THERESE T. SOMBRERO
Researchers
Contact Number: 09938420655

Noted by:

JAMES NICOLO S.J. SIAS, MLIS
Thesis Adviser

Approve by:

CHALLIZ D. OMOROG, DIT
Dean, CCS

Appendix B
SURVEY QUESTIONNAIRE

Republic of the Philippines
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Dear Respondents,

Good day!

We, the researchers of Bachelor of Library and Information Science (BLIS), conducting a study entitled **"Technology Integration Promoting Reading Literacy in Iriga City Public Library"**. In line with this we would like to ask for a few minutes of your time to answer the following questions as truthful as possible. Your participation is highly appreciated and great help in our study. Rest assured that all personal information and necessary data from this survey will be treated with outmost confidentiality.

Thank you so much and God Bless.

QUESTIONNAIRE:

I. PROFILE OF THE RESPONDENTS

Name**(optional)**: ______________________________ Age: ______

Gender: ____ Male _____Female _____ others

Type of Library User: ____ Student ____ Professionals ____ Parent ____ Scholars/Researcher ____Others

II. TECHNOLOGY INTEGRATION FOR READING LITERACY

Instructions: Which of the following technologies that you want to integrate or be available in Iriga City Public Library? Please check (/) the table using the following scale below:

4 –Strongly Agree 3 –Agree 2 –Disagree 1 – Strongly Disagree

INDICATORS	4	3	2	1
1. Electronic books - A book in electronic copy format that can be read using a computer or handheld device such as a cellphone or tablet that allows readers to download and read e-books simultaneously. Examples: Amazon kindle, Barnes and Noble, International Children's Digital Library, Free Kids Book and Story Jumper				
Other Suggestions:				
2. Audio-books - It is a recording of a book or other work being read out loud and it is can be downloaded in a digital format that allows users to listen to narrated versions of books rather than reading text, audiobooks enhance speaking and pronunciation especially for readers. Examples: Audible, Storynory, KidsRead2Kids, LibriVox, and Loyal Book				
Other Suggestions:				
3. Electronic Gadgets – It is known as the laptop, computer, tablet or mobile phone that can provide a vast amount of reading materials online or offline that has an access to educational tools that can improve the library users reading comprehension and vocabulary. Example: Laptop, Mobile phone, tablets, speaker and computer				
Other Suggestions:				
4. Educational E-Games - An interactive electronic game designed to specifically target and improve reading skills that incorporates features that can make reading practice more interactive and engaging to the readers. Examples: Doulingo, WordJong Saga, Epic, BrainGymJr-play.solve.learn!,				
Other Suggestions:				
5. Reading applications are software programs that offers a vast amount of reading materials, including phonics applications specifically designed to improve users reading skills Examples: Nabu app, hooked on phonics, reading eggs, Kids A-Z, and ReadingIQ				
Other Suggestions:				

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6. Interactive Displays - A larger version of a tablet or computer with a touchscreen technology for browsing the web where users can able to access, navigate, and interact with the content directly on the screen related to reading literacy. Examples: Television/ Smart TV, Interactive Whiteboards, Projector Display				
Other Suggestions:				
7. Online Tutoring Platforms - It is an online conferencing system that could facilitates the virtual tutoring that is specifically designed for online reading providing personalize instructions tailored to each user's specific need and reading level. Examples: Skooli, Online reading tutors, TutorMe				
Other Suggestions:				
8. ICT Facilities – A strong technological infrastructure to accommodate new integration of technology, computer workstations and digital collections. Maintaining and upgrading this will benefit the library offering of variety of digital resources and programs that will enhance reading literacy to users and promote love of reading within the community. Examples: Audio-visual room, E-laboratory, Computer Workstations				
Other Suggestions:				

III. CHALLENGES IN TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY

4 – Strongly Agree *3 – Agree* *2 – Disagree* *1 – Strongly Disagree*

Instruction: What are the challenges encountered of Iriga City Public Library in integrating technology promoting the reading literacy on terms of the following indicator?

INDICATORS	4	3	2	1
1. LIMITED ICT SKILL				
a) Untrained library staff on ICT technologies.				
b) The library doesn't have available technology for IT Staff to enhance the ICT skills				
c) There is no skills training provided for the staff on using ICT technologies.				
2. LIMITED FUND FOR TECHNOLOGY INTEGRATION				
a) Some technologies are expensive and cannot afford to the library's budget.				
b) The library prioritizes other operational needs over technology integration for reading literacy, resulting in slower progress or partial implementation of digital initiatives.				
c) The library doesn't hold any fundraising program or activities such as holding book fair, set up a donation kiosk, or cooperation in the community to raise funds.				
3. LIBRARY SPACE AND CAPACITY				
a) The library doesn't have dedicated space for new ICT equipment or digital learning areas.				
b) Not enough seats or study areas to accommodate the number of library users.				
c) Not properly organized and arranged furniture and shelves.				
4. INTERNET CONNECTIVITY				
a) The library has a slow or unreliable internet access.				

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b)	The library has one internet service provider that can provide internet connectivity to the users.				
c)	Overcrowded networks lead to poor connection to the library users.				
5. USAGE POLICIES					
a)	No existing approved usage policy on new and existing technology services and program of the library.				
b)	Unclear policy statements on how to use and manage technology resources within the library.				
c)	The use of library technology is not aligned with the established policy.				

6. LACK OF IT AND STAFF					
a)	The library has currently no applicants for IT position.				
b)	There's no technology experts' staff in the public library.				
c)	The public library's current services do not require IT support to maintain the essential library technology services.				
7. TECHNOLOGY SAFETY					
a)	Software applications are not compatible with various electronic devices own by the users.				
b)	The technologies are not covered by physical protections like cases and covers from drop and bumps.				
c)	Improper storage facility inside the library for electronic devices.				
8. TECHNOLOGY ACCESSIBILITY					
a)	The users cannot be able to access the technology if they're not in the library remotely.				
b)	Lack of orientation programs about technology accessibility.				
c)	Some library users have limited experience or knowledge about how to effectively use digital tools.				

Appendix C NON-DISCLOSURE AGREEMENT FORM

Republic of the Philippines
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NON-DISCLOSURE AGREEMENT FORM

EFFECTIVE DATE: December 23, 2023

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and **RONABETH LOQUINARIO-FIDER**, hereinafter referred to as "Panel Chairman"

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

   RONABETH LOQUINARIO-FIDER, *Panel Chairman*

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

   CHALLIZ D. OMOROG, DEAN, CCS

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:
   BLACKMANTA : **YAON, JHOEMIE A.**
   JACOB, BEVERLY G.
   SOMBRERO, MA. THERESE
   which is entitled:

   TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information. Expert would normally use with respect to its own confidential information.

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ISO 9001:2015 CERTIFIED

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COMPUTER STUDIES

7. The obligations imposed upon an Expert hereunder do not apply to information:
    a. which is or becomes publicly available without breach of this Agreement;
    b. which is already known to the Recipient prior to its disclosure hereunder;
    c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

RONABETH LOQUINARIO-FIDER
Panel Chairman

DATE: December 23, 2023

CAMARINES SUR POLYTECHNIC COLLEGES

CHALLIZ D. OMOROG, DIT
CSPC Representative

DATE: December 23, 2023

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COMPUTER STUDIES

NON-DISCLOSURE AGREEMENT FORM

EFFECTIVE DATE: December 23, 2023

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and **MILDRED A. VOLANTE,** hereinafter referred to as "Panel Member"

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

   MILDRED A. VOLANTE, *Panel Member*

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

   CHALLIZ D. OMOROG, DEAN, CCS

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:
   BLACKMANTA : **YAON, JHOEMIE A.**
   JACOB, BEVERLY G.
   SOMBRERO, MA. THERESE

   which is entitled:

   TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

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5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information. Expert would normally use with respect to its own confidential information.

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7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY: **CAMARINES SUR POLYTECHNIC COLLEGES**

MILDRED A. VOLANTE
Panel Member

CHALLIZ D. OMOROG, DIT
CSPC Representative

DATE: December 23, 2023

DATE: December 23, 2023

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NON-DISCLOSURE AGREEMENT FORM

EFFECTIVE DATE: December 23, 2023

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and **JEAN MARICON D. REGASPI,** hereinafter referred to as "Panel Member"

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

   JEAN MARICON D. REGASPI, *Panel Member*

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

   CHALLIZ D. OMOROG, DEAN, CCS

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:
   BLACKMANTA : **YAON, JHOEMIE A.**
   JACOB, BEVERLY G.
   SOMBRERO, MA. THERESE
   which is entitled:

   TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
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   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
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7. The obligations imposed upon an Expert hereunder do not apply to information:

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b. which is already known to the Recipient prior to its disclosure hereunder;
c. which is independently developed by the Expert.

8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

JEAN MARICON D. REGASPI
Panel Member

DATE: December 23, 2023

CAMARINES SUR POLYTECHNIC COLLEGES

CHARLIZ D. OMOROG, DIT
CSPC Representative

DATE: December 23, 2023

Appendix D
Joint Affidavit of Undertaking

Joint Affidavit of Undertaking

We, Jhoemie A. Ya-on, of legal age, single and a resident of Zone 3, San Vicente Bato, Camarines Sur, (2) Beverly G. Jacob, of legal age, single and a resident of Zone 1, San Miguel Nabua Camarines Sur, (3) Ma. Therese T. Sombrero, of legal age, single and a resident of Zone 2 Soroan St. La Purisima Nabua Camarines Sur, after having been sworn to accordance with Law, do hereby take oath and state:

(i) That we are officially enrolled for the thesis on the topic titled SECURITY AND SAFETY OF PRINTED MATERIALS IN CAMARINES SUR POLYTECHNIC COLLEGE LIBRARY. In the COLLEGE OF COMPUTER STUDIES OF CAMARINES SUR POLYTECHNIC COLLEGES.

(ii) That, the contents of our thesis submitted to the Camarines Sur Polytechnic Colleges, for award of BACHELOR OF LIBRARY AND INFORMATION SCIENCE. Degree is original and our own work, and is not plagiarized.

(iii) That, if, after completing this thesis, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.

(iv) That this work has not been submitted for the award of any other Degree/Diploma in any other universities/Institutes.

(v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis.

IN WITNESS WHEREOF, I have hereunto set my name this ______day of NOV 22 2024 in NABUA, CAMARINES SUR Philippines.

(1) JHOEMIE A. YA-ON
Affiant
Student ID no. C20101993

(2) BEVERLY G. JACOB
Affiant
Student ID no. C20102110

(3) MA. THERESE T. SOMBRERO
Affiant
Student ID no. C20101794

SUBSCRIBED AND SWORN TO before me this ____ day of NOV 22 2024 at NABUA, CAMARINES SUR Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No. 172 :
Page No. 36 :
Block No. VI :
Series of 2024

ATTY. FEROZZA DELIA CASTRO SIMBULAN
NOTARY PUBLIC
Commission No. IR-174 valid until 31 December 2025
Roll of Attorneys No. 87451
PTR No. 2312321 / 01-04-2024 / Nabua, Camarines Sur
IBP No. 360015 / 09-13-2023 / Camarines Sur
MCLE Compliance Admitted to the Philippine Bar in 2023
(G.B.O. No. 1, s. 2008)
2nd flr, Mequene Abe Commercial
San Francisco, Nabua, Camarines Sur, 4434

Appendix E
Project Team Assignment Form

Republic of the Philippines
CAMARINES SUR
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PROJECT TEAM ASSIGNMENTS FORM

Team Alias	**BLACK MANTA**
Course Code	**LIS 3114**
Subject Adviser/TSA	**Ms. Kay Angeline O. Broqueza**

Name and Signature	Project Role	Email Address/ Mobile Number(s)
JHOEMIE A. YA-ON	Project Head	jhoyaon@my.cspc.edu.ph 09938420655
BEVERLY G. JACOB	Information Manager	bevjacob@my.cspc.edu.ph
MA. THERESE T. SOMBRERO	Information Manager	ma.sombrero@my.cspc.edu.ph

Appendix F
Pre-proposal Title Form

Republic of the Philippines
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PRE-PROPOSAL TITLE TEMPLATE

Team Alias:	**BLACK MANTA**
Project Title:	TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY
Proponents/Researchers:	1. Jhoemie A. Ya-on 2. Beverly G. Jacob 3. Ma. Therese T. Sombrero
Scope of the Study	The aim of the research study is to investigates how technology integration can improve reading literacy among public library users in Iriga City.
Limitations of the Study:	The research study aims to determine what types of technology to integrate in the Iriga City Public Library in promoting reading literacy among library users.
Project Design/Development Plan (Software Methodology):	Program Specification
	Software Specification
	Hardware Specification
[✗] **Recommended** **KAY ANGELINE O. BROQUEZA, RL, MLIS** **Signature of SA over printed name**	[ ] **Rejected** **KAY ANGELINE O. BROQUEZA, RL, MLIS** **Signature of SA over printed name**

Appendix G
Final Project Title Form

Republic of the Philippines
CAMARINES SUR
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FINAL PROJECT TITLE FORM

Team Alias: BLACKMANTA
Proponents/Researchers:

1. Jhoemie A. Ya-on
2. Beverly G. Jacob
3. Ma. Therese T. Sombrero

Proposed Thesis Title:

TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

Submitted by: JHOEMIE A. YA-ON (Signature of Project Head over printed name) Date: ____________	Noted: KAY ANGELINE O. BROQUEZA RL, MLIS (Signature of Subject adviser over printed name) Date: ____________
Recommending Approval: JAMES NICOLO SJ. SIAS, MLIS (Signature of Adviser over printed name) Date: ____________	Approved: CHARLIZ D. OMOROG, DIT DEAN, CCS Date: ____________

Appendix H
Thesis Hearing Form (TD, POD, FOD)

Republic of the Philippines
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THESIS HEARING FORM

Date Filed: December 23, 2023 [/] Title Proposal [ ] Pre-Oral [ ] Final Oral

Date of Hearing: December 23, 2023 Time: 11:00 am Venue: LRDS

Department: **College of Computer Studies**

Research Title:

TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

Proponent/s:

YAON, JHOEMIE
JACOB, BEVERLY G.
SOMBRERO, MA THERESE

Recommended by:

JAMES NICOLO SJ. SIAS, MLIS
Thesis Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

JEAN MARICON D. REGASPI, LPT, MAED, GC
PANEL MEMBER 1

MILDRED A. VOLANTE, RL, MLIS
PANEL MEMBER 2

RONABETH L. FIDER, RL, MLIS
PANEL CHAIR

NOTED BY:

KAY ANGELINE O. BROQUEZA, RL, MLIS
SUBJECT ADVISER

APPROVED BY:

CHALLIZ D. OMOROG, DIT
DEAN, CCS

Republic of the Philippines
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THESIS HEARING FORM

Date Filed: December 23, 2023 [ ] Title Proposal [ ] Pre-Oral [/] Final Defense

Date of Hearing: June 23, 2024 Time: 8:00am Venue: AVR

Department: **College of Computer Studies Research**

Title:

TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

Proponent/s:

YA-ON, JHOEMIE A.
JACOB, BEVERLY G.
SOMBRERO, MA THERESE T.

Recommended by:

JAMES NICOLO SJ. SIAS, MLIS
Thesis Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

JEAN MARICON D. REGASPI, LPT, MAED, GC
PANEL MEMBER 1

MILDRED A. VOLANTE, RL, MLIS
PANEL MEMBER 2

RONABETH L. FIDER, RL, MLS
PANEL CHAIR

NOTED BY:

KAY ANGELINE O. BROQUEZA, RL, MLIS
SUBJECT ADVISER

APPROVED BY:

CHALLIZ D. OMOROG, DIT
DEAN, CCS

Republic of the Philippines
CAMARINES SUR
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THESIS HEARING FORM

Date Filed: May 31, 2024 [ ] Title Proposal [/] Pre-Oral [ ] Final Oral

Date of Hearing: June 04, 2024 Time: 8:00 am Venue: LRDS

Department: **College of Computer Studies**

Research Title:

TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

Proponent/s:

YAON, JHOEMIE
JACOB, BEVERLY G.
SOMBRERO, MA THERESE

Recommended by:

JAMES NICOLO SJ. SIAS, MLIS
Thesis Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

JEAN MARICON D. REGASPI, LPT, MAED, GC
PANEL MEMBER 1

MILDRED A. VOLANTE, RL, MLIS
PANEL MEMBER 2

RONABETH L. FIDER, RL, MLIS
PANEL CHAIR

NOTED BY:

KAY ANGELINE O. BROQUEZA, RL, MLIS
SUBJECT ADVISER

APPROVED BY:

CHALLIZ D. OMOROG, DIT
DEAN, CCS

Appendix I Role Acceptance Form

A. Adviser

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ROLE ACCEPTANCE FORM
College of Computer Studies

Date: November 6, 2023

To: **JAMES NICOLO SJ. SIAS**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF LIBRARY AND INFORMATION SCIENCE, are currently enrolled in Thesis 1.

We are writing to humbly request your service and expertise to serve as our Adviser for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Blackmanta

To: **DR. CHALLIZ D. OMOROG**
Dean, CCS

This formally signifies that I ACCEPT the request to serve as Adviser of the team **Blackmanta**

As Adviser, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis.

Conformed:

JAMES NICOLO SJ. SIAS, RL, MLIS
Name and Signature

B. Statistician

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COLLEGE of
COMPUTER STUDIES

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: November 6, 2023

To: **ENGR. ROWEL DOMINGO N. BERMAL**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF LIBRARY AND INFORMATION SCIENCE, are currently enrolled in Thesis 1.

We are writing to humbly request your service and expertise to serve as our Statistician for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Blackmanta

To: **DR. CHALLIZ D. OMOROG**
Dean, CCS

This formally signifies that I ACCEPT the request to serve as Statistician of the team **Blackmanta**

As Statistician, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students andensure the success of the thesis.

Conformed:

ENGR. ROWEL DOMINGO N. BERMAL
Name and Signature

C. Grammarian

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COLLEGE of
COMPUTER STUDIES

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: November 6, 2023

To: **ELAINE MAE G. TORMES**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in BACHELOR OF LIBRARY AND INFORMATION SCIENCE, are currently enrolled in Thesis 1.

We are writing to humbly request your service and expertise to serve as our Grammarian for our thesis. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Blackmanta

To: **DR. CHALLIZ D. OMOROG**
Dean, CCS

This formally signifies that I ACCEPT the request to serve as Grammarian of the team **Blackmanta**

As Grammarian, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Thesis 1 until Thesis 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students adensure the success of the thesis.

Conformed:

ELAINE MAE G. TORMES
Name and Signature

Appendix J
Panel RSC (TD, POD, FOD)

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PROJECT TITLE : **Technology Integration in Children Services in Iriga City Public Library**

TEAM ALIAS : **BLACK MANTRA (Group 3)**

DATE : **DECEMBER 23, 2023 / 12:00 PM** [/] TD [ ] POD [ ] Final [ ] For Bluebook

SECRETARY : **JENIFFER R. SEVILLA**

CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	Title Page	**Original:** Technology Integration in Children Services in Iriga City Public Library **Suggested:** Technology Integration in Promoting Reading Literacy in Iriga City Public Library	
1		**Background of the Study** • General to Specific **Statement of the Problem:** **SOP #1** • Demographic Profile .1 Type of users (students, parents…) .2 Gender .3 Educational background **SOP #2** • What are the technologies integrated in promoting reading literacy in ICPL? - Be specific on technologies used in reading literacy. **SOP #3** • What are the challenges encountered by CIPL in integrating technology in promoting reading literacy? **SOP #4** • What strategies can be proposed to enhance technology integration in reading literacy? **Objectives:** • Align to revised SOP. **Significance of the Study:** • Refer to the format used by CSPC. **Scope and Limitations:**	

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		• Indicate the focus group.	
3		**Framework** • Operational – refer to the figure on your manuscript c/o ma'am Con. • Theoretical – use **Technology Acceptance Model**	
		Questionnaire: • Align indicator/s to your revised SOPs.	
		General: • Organize your pagination	

Noted:

JEAN MARICON D. REGASPI, LPT, MAED, GC
PANEL MEMBER 1

MILDRED A. VOLANTE, RL, MLIS
PANEL MEMBER 2

RONABETH LOQUINARIO-FIDER, MLS
PANEL CHAIRMAN

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PROJECT TITLE : Technology Integration in Promoting Reading Literacy in Iriga City Public Library

TEAM ALIAS : BLACK MANTRA (Group 1)

DATE : JUNE 04, 2024 / 9:00 AM [ ] TD [/] POD [ ] Final [ ] For Bluebook

SECRETARY : JENIFFER R. SEVILLA

CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS,AND COMMENTS (RSC)	ACTION TAKEN
	Abstract	• Follow **IMRAD** format (introduction, Methods, Results, and Discussion)	
1		**Significance of the Study:** • Include the significance of the study for the specific group for which the study is intended. **Project Dictionary:** • Put additional terminologies	
2		**Gap Bridged of the Study** • Delete Paragraph #1(not related), and add another that is related **Synthesis of the State-of-the-Art** • Put additional information	
3		**Conceptual Framework** • Use line, not arrow; no relationship **Statistical Treatment of Data** • Add **rank** and **percentage** **Likert Scale** • Do not put on table • Straightforward description	
4		**Table for the Profile** • Make one, composite table **Table #2** • Technologies that can be integrated **Table #4** • Revise the explanation • Discuss the result • Composite table (challenges)	
5		**Conclusion** • Relate to the findings	
Output		• Contingency Plan to be proposed for the Iriga City Public Library	
		General: • Organize your table of contents and pagination and follow the CSPC format. • Use proper tenses	

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		• Follow ACM Citation Format	

Noted:

JEAN MARICON D. LEGASPI, LPT, MAED, GC
PANEL MEMBER 1

MILDRED A. VOLANTE, RL, MLIS
PANEL MEMBER 2

RONABETH LOQUINARIO-FIDER, MLIS
PANEL CHAIRMAN

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PROJECT TITLE : **Technology Integration in Promoting Reading Literacy in Iriga City Public Library**

TEAM ALIAS : **BLACK MANTRA**

DATE : **JUNE 22, 2024** [ ] TD [ ] POD [/] Final [ ] For Bluebook

SECRETARY : **JENIFFER R. SEVILLA**

CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS,AND COMMENTS (RSC)	ACTION TAKEN
	Abstract	• Follow **IMRAD** format (introduction, Methods, Results, and Discussion)	
3		**Likert Scale** • Remove label "Table 1" • Verbal interpretation shall be used in the discussion of data	
4		**Table for the Demographic Profile** • Make one, composite table **Table #2** • Rank – in number (1,2,3…) **Table #4** • Composite table (challenges) • Rank – in number (1,2,3…) **On discussing the Table** • Interpret well, and make some implications to the findings of the study -	
5		**Findings** • Present the results **Conclusion** • Make it concise **Recommendation** • Recommendation #1 shall be in response to SOP #1, and so on. • Include the fact that the study will be used as the basis for the renovation of ICPL. **1 to 1 Based** • Conclusion and Recommendation should be	

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		parallel to the Results, with respect to the order of number.	
Output		• **Contingency Plan** - Put a short introduction - Append • Include in Chapter 4 the discussion/ introductory part of the output	
		General: • Organize your table of contents, margin, and pagination and follow the CSPC format. • Use proper tenses • Follow ACM Citation Format	

Noted:

JEAN MARICON D. REGASPI, LPT, MAED, GC
PANEL MEMBER 1

MILDRED A. VOLANTE, RL, MLIS
PANEL MEMBER 2

RONABETH LOQUINARIO-FIDER, MLIS
PANEL CHAIRMAN

Appendix K
Consultation Log Form

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CONSULTATION LOG FORM

Thesis/Capstone Project Title	TECHNOLOGY INTEGRATION PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY						
Proponents:	1. Jhoemie A. Ya-on			2. Beverly G. Jacob			
	3. Ma.Therese T. Sombrero						
Alias:	Blackmanta						
Total # of Modules			As approved by the CAPSA/TSA				
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Module Partially Implemented	Running Score	Percentage	Project Manager's Signature	TA/TSA/CAPA/CAPSA

	Remarks
Chapter 1	• From Sir James Nicolo SJ. Sias - revise your chapter 1, must be in line in your topic, make sure that the references you use is reliable, and check your citations. It is going to be a ACM citation.
	• From Sir James Nicolo SJ. Sias - For profile of respondents in statement of the problem 1, inquire and classify the type of users that enters the public library in Iriga city.
	• From Sir James Nicolo SJ. Sias - Just put blank after the Age and let the respondents put them and while answering the survey questionnaire as it is a necessary data for the research paper.
	• From Sir James Nicolo SJ. Sias - In statement of the problem 3, observe and list down all the possible challenges encountered of the Iriga city public library in integrating technology.
Chapter 2	• From Sir James Nicolo SJ. Sias - You need to have at least 30 Related literatures, that is related to your study. The RRL must be in line in your topical outline.
	• From Sir James Nicolo SJ. Sias – Find Related Literature that is about the General Public Libraries and Advanced PH Libraries.
Chapter 3	
Deadline:	

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CONSULTATION LOGS FORM

Thesis/Capstone Project Title	TECHNOLOGY INTEGRATION PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY						
Proponents:	1. Jhoemie A. Ya-on			2. Beverly G. Jacob			
	3. Ma.Therese T. Sombrero						
Alias:	Blackmanta						
Total # of Modules			As approved by the CAPSA/TSA				
PROTOTYPE	DT of Consultation	# of Modules Fully Implemented	# of Module Partially Implemented	Running Score	Percentage	Project Manager's Signature	TA/TSA/CAPA/CAPSA

	Remarks
Chapter 4	✓ From Sir James Nicolo SJ. Sias – Analyze, explain, interpret give a supporting detail, and supporting literature.
	✓ From Sir James Nicolo SJ. Sias – Each table it should have a supporting literature that is connected to the table.
	✓ From Sir James Nicolo SJ. Sias – Double check each table, many there's a mistake.
	✓ From Sir James Nicolo SJ. Sias – In interpreting each table, it should be connected in every table, or what are the results trying to explain.
	✓ From Sir James Nicolo SJ. Sias – Check the grammar, term used, punctuation marks and capitalization of term after the period.
	✓ From Sir James Nicolo SJ. Sias - Double check the result percentage when interpreting the gathered data.
	✓ From Sir James Nicolo SJ. Sias – Make sure the supporting details and the interpretations are correct.
	✓ From Sir James Nicolo SJ. Sias – Combine the 3 tables in the respondents' profile, into 1 table.
Chapter 5	✓ From Sir James Nicolo SJ. Sias – In chapter 5, it is the summarize explanation of the chapter 4.
	✓ From Sir James Nicolo SJ. Sias - In doing the recommendation, you will think or what is the primary recommendations of your group to help or assist the problem in technology integration especially in Iriga City Public Library.
	✓ From Sir James Nicolo SJ. Sias – In the last number of the recommendation, put introduction for the proposed output.
	✓ From Sir James Nicolo SJ. Sias – Findings are the summary in each interpreted table in chapter 4. The conclusion is the summary of the findings.
	✓ From Sir James Nicolo SJ. Sias - Check the recommendation and base it on your statement of the problem.

Appendix L
Editor's Certification

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CERTIFICATION

This is to certify that the undersigned has reviewed and went through all pages of the thesis entitled "**TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY**" as against the set of structural rules that govern research writing in account with the composition of sentences, phrases, and words in the English Language.

Signed:

MS. ELAINE MAE G. TORMES
Grammarian

Date Signed: November 25, 2024

Appendix M
Secretary's Certification

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CERTIFICATION

This is to certify that the undersigned has provided true and correct recommendations, suggestions, and comments unanimously agreed and approved by the panel of examiners during the oral examination of the [thesis/capstone project] entitled **"TECHNOLOGY INTEGRATION PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY"** prepared and submitted by Jhoemie A. Ya-on, Beverly G. Jacob and Ma. Therese T. Sombrero and that the same have not been amended, modified or obliterated.

Signed:

MS. JENIFFER R. SEVILLA
Secretary

Date Signed: November 22, 2024

Appendix N
Statistician's Certificate

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C E R T I F I C A T I O N

This is to certify that the thesis entitled **"TECHNOLOGY INTEGRATION PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY"** prepared and submitted by Jhoemie A. Ya-on, Beverly G. Jacob and Ma. Therese T. Sombrero has been statistically reviewed by the undersigned.

Signed:

MR. ROWEL D.BERMAL
Statistician

Date Signed: November 23, 2024

Appendix O

Certificate of Plagiarism Check

REPUBLIC OF THE PHILIPPINES
CAMARINES SUR POLYTECHNIC COLLEGES
Nabua, Camarines Sur

OFFICE OF THE DIRECTOR
CENTER FOR RESEARCH AND DEVELOPMENT

/cspc.crd
research@cspc.edu.ph

CERTIFICATION

This certifies that the study entitled **TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY** authored by **JHOEMIE A. YA-ON, BEVERLY G. JACOB AND MA. THERESE T. SOMBRERO** of **Bachelor of Library Information Science** was submitted and undergone to the Camarines Sur Polytechnic Colleges-Center for Research and Development (CSPC-CRD) anti-plagiarism checking on December 4, 2024.

ENGR. HAROLD JAN R. TERANO, Ph.D.
Director, Center for Research & Development

Research Office Telefax. No. (054) 288-4519 / Tel. No. (054) 288-4425 loc 152 / e-mail Address: **research@cspc.edu.ph**

TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

ORIGINALITY REPORT

7%	6%	3%	%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

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2	harbinengineeringjournal.com Internet Source	<1%
3	pdfcoffee.com Internet Source	<1%
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5	"Regional Conference on Science, Technology and Social Sciences (RCSTSS 2014)", Springer Science and Business Media LLC, 2016 Publication	<1%
6	journals.sagepub.com Internet Source	<1%
7	Sai Kiran Oruganti, Dimitrios A Karras, Srinesh Singh Thakur. "Advancing Sustainable Science and Technology for a Resilient Future", CRC Press, 2024 Publication	<1%

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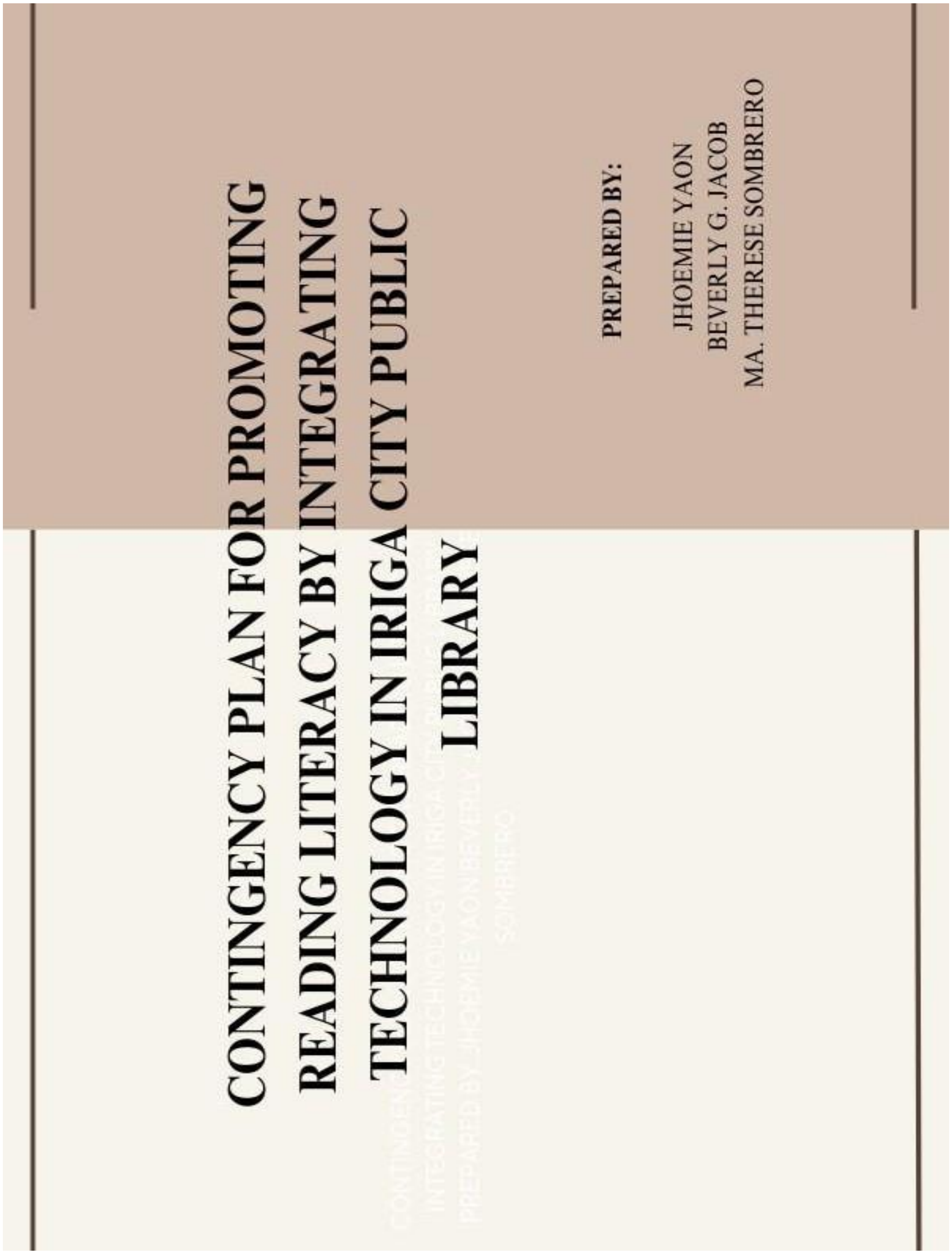
Exclude quotes	On	Exclude matches	Off
Exclude bibliography	On		

Appendix P
Proposal Output


CONTINGENCY PLAN FOR PROMOTING READING LITERACY BY INTEGRATING TECHNOLOGY IN IRIGA CITY PUBLIC LIBRARY

PREPARED BY:

JHOEMIE YAON
BEVERLY G. JACOB
MA. THERESE SOMBRERO

OBJECTIVES

- To develop a comprehensive plan that outlines strategies and procedures for mitigating the impact of technological failures or disruptions on promoting reading literacy programs in Iriga City Public Library, ensuring the continuity of essential services and minimizing disruptions to library patrons.

- To enhance the reading skills of various community members by incorporating technology into the library, meeting their diverse needs and promoting literacy.

- To addressed the challenges of integrating technology in the Iriga City Public Library, providing necessary support to users to improve their reading abilities.

- Foster reading literacy in the Iriga City Public Library through the adoption of multiple technologies, ensuring that the library serves as a key educational resource for personal and academic growth of community users.

RATIONALE

- The integration of technology in public libraries is increasingly necessary to promote reading and ensure equitable access to information. At the Iriga City Public Library, the adoption of digital tools and resources can significantly improve the reading experience, making it more engaging and accessible to all users. However, challenges such as limited staff IT skills, insufficient funding, and inadequate internet connectivity can hinder the effective implementation of these technologies. Therefore, a well-structured contingency plan is essential to address these challenges and ensure the successful integration of technology to promote reading.

Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
LIMITED ICT SKILL OF LIBRARY STAFF IN THE IRIGA CITY PUBLIC LIBRARY	• Staff struggling to assist patrons with digital resources and it challenge them to manage digital catalogues, databases, and automated systems. • Limiting the patrons the ability to utilize digital resources due to inadequate support of staff in accessing them • The library is vulnerable to digital threats	• Provide training and interventions for library staffs. • Partner with local colleges or universities to provide ICT training and resources. • Organize regular workshops and refresher courses on emerging technologies and cybersecurity practices.	• Library head • Local Government Unit • Training Coordinator, External ICT Trainers.	• Improved staff proficiency in ICT, enabling better assistance to patrons and efficient management of digital resources. • Access to expert knowledge and potential for ongoing collaboration. • Continuous improvement in ICT skills and up-to-date knowledge on cybersecurity.	Within 6 Months, Starting January to June 2025

Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
LIMITED FUNDING FOR TECHNOLOGY INTEGRATION	• Hindering the public library from procuring new digital resources. • The iriga city public library might struggle to maintain or upgrade their existing technology. • It hinders the development and implementation of digital literacy programs to help patrons navigate and utilize digital resources effectively.	• Apply for grants, seek donations, and explore partnerships with local businesses and educational institutions. • Utilize open-source software and free digital resources to supplement existing technology. • Launch community campaigns to raise awareness about the library's needs and the benefits of technology in reading.	• Library Head • IT Department • Library Staff • Community Outreach Coordinator	• Increased funding to acquire new technologies and resources. • Enhanced digital offerings without significant financial investment. • Increased community support and potential funding through donations and local.	Within 12 Months, Starting January to December 2025

Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
LIBRARY SPACE AND CAPACITY IN THE IRIGA CITY PUBLIC LIBRARY	• Overcrowding due to insufficient space. • It restricts the ability to host programs, workshops, and community events. • Limits the library to expand its collection of books, digital resources, and other materials	• Rearrange and redesign existing spaces to maximize usability and efficiency. • Partner with local schools, community centers, and businesses to use their spaces for library programs and events. • Focus on expanding digital collections and online services to reduce the need for physical space.	• Library Head • IT Department • Library Staff • Community Outreach Coordinator	• Improved spaces utilization, enhanced user experience. • Increased capacity for programs and events. • Enhance access to resources without the need for additional physical space.	Within 6 Months, Starting January to June 2025

Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
IRIGA CITY PUBLIC LIBRARY INTERNET CONNECTIVITY	• Struggling to access e-books, online databases, and other digital resources, limiting their ability to utilize the library's full range of offerings. • It impede the library's ability to offer digital literacy programs, which are essential for helping patrons develop necessary digital skills • It might frustrate Patrons with slow or unreliable internet connections, leading with the decreased usage of library services and resources.	• Invest in upgrading the library's internet infrastructure to ensure high-speed, reliable connectivity. • It might frustrate Patrons with slow or unreliable internet connections, leading with the decreased usage of library services and resources. • Establish a program to lend Wi-Fi hotspots to patrons, providing internet access outside the library.	• Library Head • Local Government • Fundraising Coordinator • Community Outreach Coordinator	• This Improved internet speed and reliability, enhancing access to online resources and digital literacy programs. • Enhanced internet connectivity through shared resources, fostering community collaboration. • Secured funding to support long-term internet connectivity improvements.	Within 12 Months, Starting January to December 2025

Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
TECHNOLOGY SAFETY IN USING TECHNOLOGIES	• The technologies are not covered by physical protections like cases and covers from drop and bumps. • Improper storage facility inside the library for electronic devices. • Improper installation or use of technology equipment that can lead to serious physical dangers to users, such as electrical fires or injuries from unstable setups.	• Implement safety measures for technology resources. • Establish clear guidelines for handling equipment damage, including the repair costs, and replacement procedures. • Ensure safe installation and maintenance of equipment	• Library Head • Library Staff • IT Coordinator • IT Staff	• Conduct a comprehensive security audit and implement necessary measures. • Develop and distribute educational materials on technology safety to patrons. • Conduct regular safety inspections and training sessions	Within 6 Months, Starting January to June 2025

Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
TECHNOLOGY ACCESSIBILITY IN TECHNOLOGY	• The users cannot be able to access the technology if they're not in the library remotely. • Library users may not effectively utilize technological resources without adequate training and support. This could lead to investments being wasted and opportunities to improve reading literacy and other educational results being overlooked. • Struggling to use or access technology can cause frustration for library patrons, which may result in decreased visits to the library and participation in reading initiatives.	• Implement workshops on using library databases, searching online effectively, and even downloading eBooks. • Provide user-friendly technology access. • Offer continuous support and assistance	• Municipal Mayor • Library Head • IT Coordinator • IT Staff • Library Staff	• Focus on expanding remote access to resources and enhancing user comfort with technology. • Install user-friendly interfaces and assistive technologies • Set up help desks and support services for technology users	Within 6 Months, Starting January to June 2025

Appendix Q
ACM Format

TECHNOLOGY INTEGRATION IN PROMOTING READING LITERACY IN IRIGA CITY PUBLIC LIBRARY

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ABSTRACT
This study investigates the integration of technology to enhance reading literacy at the Iriga City Public Library. It identifies specific technologies to improve reading literacy and addresses the challenges of incorporating digital tools into a traditional library setting. Using a quantitative, descriptive design and the Technology Acceptance Model, data was gathered from 355 respondents, including students, professionals, parents, PWDs, pregnant women, senior citizens, and out-of-school youth. The findings revealed that most respondents were students who preferred electronic gadgets for reading. However, challenges such as accessibility to digital tools were noted. To address these issues, a contingency plan was proposed focusing on enhancing user experience, identifying risks, providing solutions, having backup plans, and promoting adaptability and resilience. The goal is to ensure continuous service, protect resources, and improve reading literacy through effective technology integration, thereby fostering a culture of reading and enhancing literacy rates among library patrons.

KEYWORDS: Reading literacy, Technology, Public Library, Iriga City

1. INTRODUCTION
Libraries are essential resources for education, research, and information. Advancement in information and Communication Technology (ICT) have led to the development of mobile library applications and digital libraries, which users can access from various locations using computers and the internet [1]. Public Libraries, created and overseen by community members, play a significant social role by providing knowledge and opportunities, preserving cultural heritage, and promoting public awareness [2]. Effective technology design and pedagogical strategies are crucial for teaching reading comprehension. Well-designed digital books and student-centered technology-based activities are linked to improved reading performance [3]. Technology integration involves using digital tools to enhance educational environments, including classrooms and libraries [4]. Iriga City Public Library, serving a community of 114,457 people, faces challenges in integrating new technologies due to limited funding. This study investigates these challenges and strategies for promoting reading literacy through technology [5]. The goal is to develop strategies that maximize technology integration, supporting the library in becoming a leader in inclusive literacy and creating a more empowered, literate community.

2. RESEARCH DESIGN
The researchers used quantitative data and a descriptive method to overview the current technological integration promoting reading literacy at the Iriga City Public Library. This method, as noted by Pritha Bhandari [2023], is crucial for accurately defining a

situation, population, or phenomena. The descriptive method design involved purposive sampling to select participants who visit the library and have experienced the technology integration initiatives. The study aims to improve library services, ensure equitable access to technology, and support community users in their learning and development in the digital age [6].

3. METHODOLOGY

This chapter gives an overview of the research methods and procedures applied in this study. The method or choice of respondents to be used for the analysis and interpretation of responses from participants in this field is also covered, as well as data collection instruments and documentary analyses that will be used during the study.

3.1. Research method

The researchers conducted a study using quantitative data to evaluate the technological integration that promotes reading literacy at the Iriga City Public Library. Data was collected through survey questionnaires to measure library usage, habits, and attitudes towards technology. A purposive sampling technique was employed to select participants who have experienced the library's technology initiatives. The study aims to enhance library services, ensure equitable access to technology, and support community learning and development in the digital age.

3.2. Theoretical framework

In Robert Allen's research, The Technology Acceptance Model (TAM) by Davis (1989) is used to explain technology integration at the Iriga City Public Library to promote reading literacy [7]. The study focuses on perceived usefulness, defined as the intention to integrate and the perception of using technologies. It posits that technology resources can make it easier for patrons to access and use library services. The research will examine how perceived usefulness, as influenced by TAM, affects patrons' adoption and usage of technology-based library resources.

3.3. Operational framework

As shown in Figure 1, The Technology Acceptance Model (TAM) operational framework illustrates the relationship between variables in the context of the Iriga City Public Library. The independent variables identified are the library itself and technology integration. The dependent variables are the profile of library users, strategies for technology adoption, and challenges faced during implementation. This framework highlights how factors such as education level and familiarity with technology significantly impact users' perception of the usefulness of digital reading tools. Tailoring these tools to meet diverse user needs can enhance their perceived usefulness, while strategies emphasizing practical benefits can boost their adoption. Overcoming challenges like limited technology access and resistance to change by showcasing the benefits of digital reading tools can further improve reading literacy among the library's users.

3.4. Data gathering tools and procedures

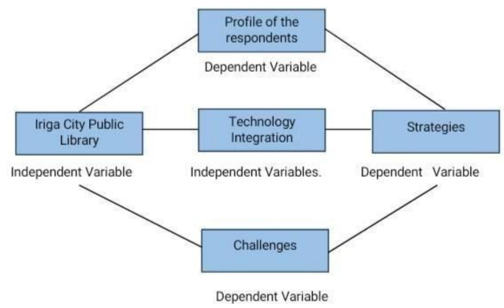


Figure 1: Operational Framework of the study

To gather the required data for the Iriga City Public Library study, researchers used a survey questionnaire checklist as the primary data collection tool. The questionnaire, developed after reviewing related literature, consisted of three parts: respondent profiles (age, gender, type of library users), To prepare the questionnaire, the researcher gathered and reviewed related literature and studies to construct the questions. The questionnaire is composed of Three (3) parts. technologies for integration, and challenges in integrating technology to promote reading literacy. Before being distributed, the developed checklist questionnaire was reviewed by the panel, adviser, instructor, and consultant. The questionnaire was reviewed, and a practice run was performed. It was handed to 20 random users from the Iriga City Public Library. The questionnaire underwent review by a panel, adviser, instructor, and consultant, followed by a trial with 20 random library users. Once approved, it was distributed to 355 respondents, who provided consent before participation. To assist the analysis process, the researchers gave the survey questionnaire results to the statistician informs the researcher that he's done computing the results, utilizing appropriate treatment. The collected data was reviewed, tabulated, and analyzed, ensuring confidentiality. For analysis, the results were given to a statistician to ensure accuracy and thoroughness in data treatment.

3.5. Statistical treatment of data

The statistical tools used by the researchers are the Slovin's formula to determine the sample size of the respondents in the Iriga City Public Library community users. After collecting responses from the questionnaire, data processing and analysis were performed to derive meaningful results. The following statistical techniques were employed: The percentage technique was used to verify the calculated percentage equivalent of the collected responses, determining the percentage of respondents who answered each question. Frequency measured how often something occurred or was repeated over a particular period. interpretations of the technologies that could be integrated into the Iriga City Public Library and the challenges faced in integrating technology to promote reading literacy. The Likert Scale assessed opinions and determined respondents' levels of agreement or preference towards specific statements, utilizing a four-point scale for calculating respondent averages, verbal interpretation, and description. This scale was essential in summarizing findings regarding technologies that can enhance reading literacy at the Iriga City Public Library.

4. RESULTS AND DISCUSSION

This chapter presents, analyzes, and interprets the data collected on technology integration in promoting reading literacy in the Iriga City Public Library. The researcher applied quantitative data includes discussions on respondents' profiles, technologies that can be integrated, and challenges encountered in technology integration to enhance reading literacy. To clarify, the information is displayed in a table format

4.1. Respondents profile

Table 1. Respondent's Profile

Profile		FREQUENCY	PERCENTAGE
Age	10-20	90	25.50
	21-30	**165**	**46.74**
	31-40	40	11.33
	41-50	36	10.20
	51 Above	22	6.23
Gender	Male	134	37.96
	Female	**219**	**62.03**
Type of Library User's	**Student**	**239**	**67.70**
	Professional	21	5.94
	Parent	44	12.46
	Others	49	13.88
Total Number of Respondents = 353			

Table 1 shows the frequency distribution and percentage of users' profiles at the Iriga City Public Library based on age and gender. Parishmita Hazarika (2021) highlighted that libraries provide critical environments for solitude, education, and social interaction, particularly for women. Out of 353 respondents, the largest age group is 21 to 30 years old (46.74%), mainly students using the library for research and assignments. In terms of gender, females make up the majority of users (62.03%), indicating that more women frequent the library due to the opportunities it offers [8]. The data also shows that 67.70% of library users are students, underscoring the library's role in supporting educational needs. Maria Juanita R. Macapagal (2023) emphasized the importance of libraries in providing essential resources and fostering a culture of reading and lifelong learning among students. She suggested improvements such as increasing library funding, updating facilities, and enhancing digital resources to better serve students' needs, presenting a comprehensive view of the strengths and areas for growth within the Philippine public library system [9].

4.2. Technologies that can be integrated in promoting reading literacy in the Iriga City Public Library.

Table 2. Technologies that can be integrated in promoting reading literacy in the Iriga City Public Library

Indicators	Weighted mean	Verbal Interpretation	Ranking
Electronic books	3.98	Strongly Agree	2
Audio-books	3.96	Strongly Agree	5
Electronic Gadgets	**3.99**	Strongly Agree	**1**
Educational E-games	3.97	Strongly Agree	4
Reading applications	3.96	Strongly Agree	5
Interactive Displays	3.92	Strongly Agree	7
Online Tutoring Platforms	3.91	Strongly Agree	8
ICT Facilities	3.98	Strongly Agree	2
Average Weighted Mean	**3.96**	Strongly Agree	

Table 2 shows an overall average weighted mean of 3.96, indicating a strong agreement among the user community of the Iriga City Public Library about integrating technologies to enhance library services and programs. This integration aims to improve access to diverse information, foster reading literacy, and promote an inclusive learning environment. The technologies that can be integrated are ranked, with electronic gadgets (weighted mean of 3.99) being the highest, followed by ICT facilities and electronic books (3.98), electronic e-games (3.97), reading applications and audio books (3.96), interactive displays (3.92), and online tutoring platforms (3.91). These findings align with the study by Marchenko et al. (2021), highlighting the benefits of technology integration for enhancing reading literacy, such as improved access to educational materials and tools for spelling, pronunciation, vocabulary, and sentence construction [10]. This technology integration promotes digitization and benefits the younger generation.

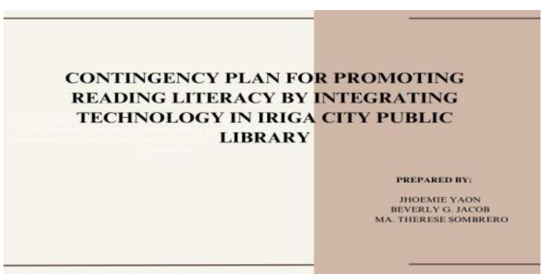
4.3. Challenges encountered of Iriga City Public Library in technology integration promoting reading literacy along the indicators

Table 3. Challenges encountered of Iriga City Public Library in technology integration in promoting reading literacy along the indicators.

Indicators	Weighted mean	Verbal Interpretation	Ranking
Limited ICT Skills	3.74	Strongly Agree	8
Limited Fund for Technology Integration	3.90	Strongly Agree	2
Library Space and Capacity	3.80	Strongly Agree	6
Internet Connectivity	3.86	Strongly Agree	3
Usage Policies	3.79	Strongly Agree	7
Lack of IT Staff	3.81	Strongly Agree	5
Technology Safety	3.83	Strongly Agree	4
Technology Accessibility	**3.92**	**Strongly Agree**	**1**
Average Weighted Mean	**3.83**	**Strongly Agree**	

Table 3 shows an average weighted mean of 3.83, indicating strong agreement among users of the Iriga City Public Library on the challenges associated with integrating technologies to promote reading literacy. Users recognize the need for technological innovation but also acknowledge difficulties in the transition. Key challenges identified include technology accessibility (3.92), limited funding (3.90), internet connectivity (3.86), technology safety (3.83), lack of IoT staff (3.81), library space and capacity (3.80), usage policies (3.79), and limited ICT skills (3.74). Technology accessibility, being the most critical issue, affects users' ability to engage with digital reading tools effectively. The study by Kuri Gurumurthy and Padmamma S. (2023) supports these findings, highlighting that ICT improves resource accessibility and library service efficiency, which are crucial for fostering reading literacy. Ensuring easy access to technology not only enhances user satisfaction but also promotes an inclusive and effective library experience, essential for improving reading literacy among diverse user groups [11].

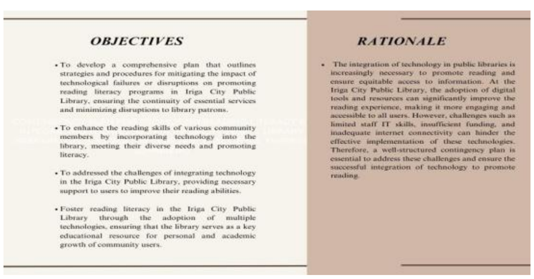
4.5 Proposed output to promote reading literacy through technology integration in Iriga City Public Library.



CONTINGENCY PLAN FOR PROMOTING READING LITERACY BY INTEGRATING TECHNOLOGY IN IRIGA CITY PUBLIC LIBRARY

PREPARED BY:

JHOEMIE YAON
BEVERLY G. JACOB
MA. THERESE SOMBRERO

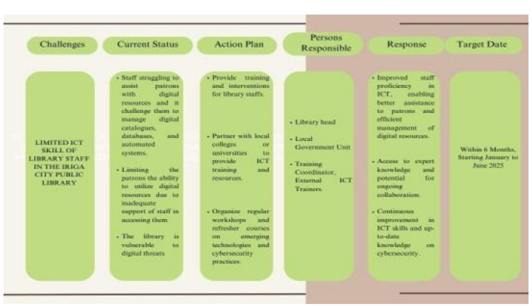


OBJECTIVES

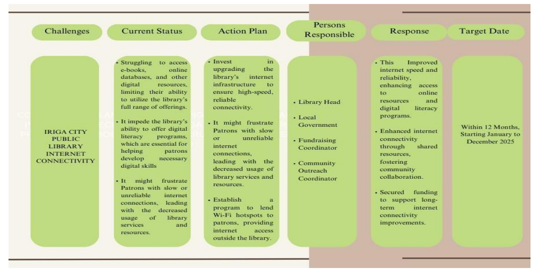
- To develop a comprehensive plan that outlines strategies and procedures for mitigating the impact of technological failures or disruptions on promoting reading literacy programs in Iriga City Public Library, ensuring the continuity of essential services and minimizing disruptions to library patrons.
- To enhance the reading skills of various community members by incorporating technology into the library, meeting their diverse needs and promoting literacy.
- To addressed the challenges of integrating technology in the Iriga City Public Library, providing necessary support to users to improve their reading abilities.
- Foster reading literacy in the Iriga City Public Library through the adoption of multiple technologies, ensuring that the library serves as a key educational resource for personal and academic growth of community users.

RATIONALE

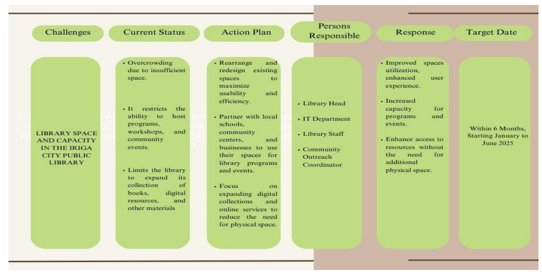
- The integration of technology in public libraries is increasingly necessary to promote reading and ensure equitable access to information. At the Iriga City Public Library, the adoption of digital tools and resources can significantly improve the reading experience, making it more engaging and accessible to all users. However, challenges such as limited staff IT skills, insufficient funding, and inadequate internet connectivity can hinder the effective implementation of these technologies. Therefore, a well-structured contingency plan is essential to address these challenges and ensure the successful integration of technology to promote reading.



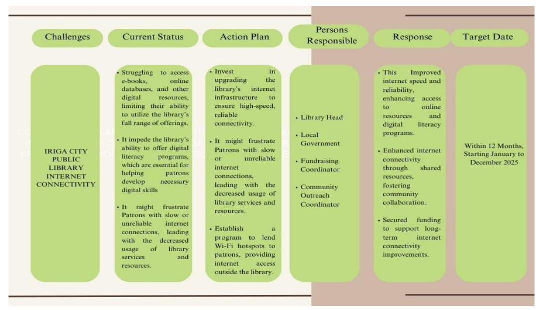
Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
LIMITED ICT SKILL OF LIBRARY STAFF IN THE IRIGA CITY PUBLIC LIBRARY	• Staff struggling to assist patrons with digital resources and it challenge them to manage digital catalogues, databases, and automated systems. • Limiting the patrons the ability to utilize digital resources due to inadequate support of staff in accessing them. • The library is vulnerable to digital threats	• Provide training and interventions for library staffs. • Partner with local colleges or universities to provide ICT training and resources. • Organize regular workshops and refresher courses on emerging technologies and cybersecurity practices.	• Library head • Local Government Unit • Training Coordinator, External ICT Trainers.	• Improved staff proficiency in ICT, enabling better assistance to patrons and efficient management of digital resources. • Access to expert knowledge and potential for ongoing collaboration. • Continuous improvement in ICT skills and up-to-date knowledge on cybersecurity.	Within 6 Months, Starting January to June 2025



Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
IRIGA CITY PUBLIC LIBRARY INTERNET CONNECTIVITY	• Struggling to access e-books, online databases, and other digital resources, limiting their ability to utilize the library's full range of offerings. • It impede the library's ability to offer digital literacy programs, which are essential for helping patrons develop necessary digital skills • It might frustrate Patrons with slow or unreliable internet connections, leading with the decreased usage of library services and resources.	• Invest in upgrading the library's internet infrastructure to ensure high-speed, reliable connectivity. • It might frustrate Patrons with slow or unreliable internet connections, leading with the decreased usage of library services and resources. • Establish a program to lend Wi-Fi hotspots to patrons, providing internet access outside the library.	• Library Head • Local Government • Fundraising Coordinator • Community Outreach Coordinator	• This Improved internet speed and reliability, enhancing access to online resources and digital literacy programs. • Enhanced internet connectivity through shared resources, fostering community collaboration. • Secured funding to support long-term internet connectivity improvements.	Within 12 Months, Starting January to December 2025



Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
LIBRARY SPACE AND CAPACITY IN THE IRIGA CITY PUBLIC LIBRARY	• Overcrowding due to insufficient space. • It restricts the ability to host programs, workshops, and community events. • Limits the library to expand its collection of books, digital resources, and other materials	• Rearrange and redesign existing spaces to maximize usability and efficiency. • Partner with local schools, community centers, and businesses to use their spaces for library programs and events. • Focus on expanding digital collections and online services to reduce the need for physical space.	• Library Head • IT Department • Library Staff • Community Outreach Coordinator	• Improved spaces utilization, enhanced user experience. • Increased capacity for programs and events. • Enhance access to resources without the need for additional physical space.	Within 6 Months, Starting January to June 2025



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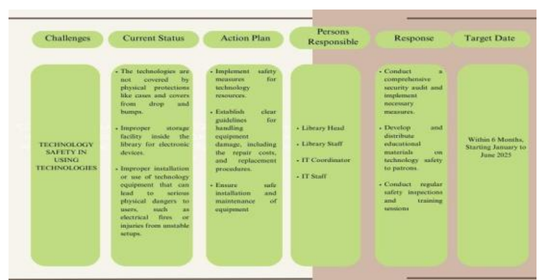

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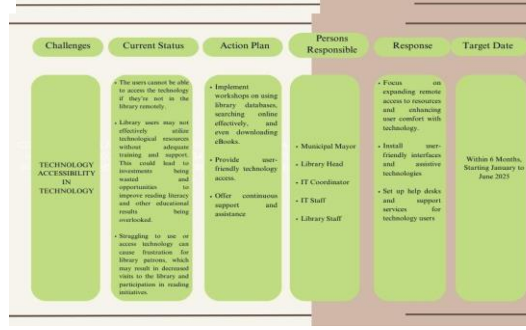
5.



Challenges	Current Status	Action Plan	Persons Responsible	Response	Target Date
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CONCLUSION AND DISCUSSION

This chapter provides a summary of the overall findings from the analyzed data, which were thoroughly investigated. It leads to conclusions and proposed

5.1. Conclusions

1. The age group of 21–30 received the highest ranking, and most of the respondents were female students' users of the Iriga City Public Library.
2. The highest-ranking indicator in determining what technologies can be integrated in Iriga City Public Library was the "Electronic Gadgets," which concludes that the Iriga City Public Library users have a strong preference for mobile devices and other portable technology to integrate for improving the library services for reading literacy and to accommodate the needs of library patrons in the community.
3. The study found that technology accessibility is the primary challenge in integrating technology to promote reading literacy at Iriga City Public Library. Users recognize the potential benefits of new technologies but may feel anxious and hesitant about using them. The researchers concluded that with proper user engagement and orientation, library patrons can effectively utilize these technologies. Proper guidance and support are essential to ensure these advancements are embraced and maximized, ultimately enhancing the reading literacy experience for all users.
4. The Iriga City Public Library's contingency plan integrates secure digital tools to enhance information access, reading literacy skills and foster lifelong learning of the public library users. The aim of the contingency plan is to utilize technology in a secure and beneficial setting, in line with the library's commitment to providing excellent services and promoting community growth.

5.2 Recommendations

Based on the findings and conclusions, the following recommendations are derived:

1. Establishment of an ICT Section or Cyber Room in the Iriga City Public Library involves creating a dedicated space equipped with computers, tablets, and high-speed internet access. This facility offers digital resources such as electronic books, audiobooks, online databases, and educational software to enhance reading skills. Strategically accessibility, the center provides technologies that support various reading and learning activities, ultimately enhancing users' reading capabilities.
2. Establish a comprehensive evaluation process to select and test high-efficiency electronic gadgets. This process should prioritize reputation, durability, to ensure compatibility with existing systems and support for. The committee should conduct thorough market research, gather user feedback, and partner with reputable suppliers to secure quality devices with the best deals and warranties. Implementing a pilot program to test gadgets before full-scale procurement will ensure that the chosen devices effectively improve reading literacy and user experience for future readers.
3. Upgrade the Wi-Fi infrastructure to support more users and create designated high-bandwidth zones. Implement a Wi-Fi management system to ensure fair access, promote online reading activities, and encourage the use of digital reading materials. This will enhance internet connectivity for all devices in the library and improve the overall user experience for future readers.
4. Hire IT staff proficient in both hardware and software digital technologies to support library users in accessing and utilizing digital reading resources. Conduct training sessions for the library's IT staff to manage and navigate digital reading infrastructure, ensuring systems are current, fully functional, and supportive of users' reading development.
5. Provide Hands-on workshops on effectively using reading tools inside the library. These should cover topics like an overview of digital resources, accessing library portals, online platforms, reading applications, and navigating various reading tools. These programs will improve users' reading skills, confidence in utilizing digital resources, and overall literacy development.
6. Developed a "Contingency Plan for Promoting Reading Literacy through Technology Integration in Iriga City Public Library" to manage and mitigate literacy in the Iriga City Public Library. It will outline challenges, proposed actions, responsible personnel, expected responses, and target dates for implementation to ensure effective technology adoption for reading literacy.

6. ACKNOWLEDGEMENTS

We extend our deepest gratitude and appreciation to the following individuals who have significantly influenced our thesis journey: Their inspiration, patience, and understanding during our trials and errors were invaluable. Our heartfelt thanks go to Dr. Challiz D. Omorog, Dean of the College of Computer Studies, Mr. James Sias, Ms. Kay Angeline O. Broqueza, RL MLIS, our Thesis Subject Adviser, Mrs. Ronabeth Fider, Mrs. Mildred A. Volante, and Ms. Jean Maricon D. Regaspi, to the panelists, Mr. Yves Aristeo Febres, Director of Learning Resources and Development Services (LRDS), Mr. Rowel Bermal, Ms. Elaine Torres, and to our loving family, friends, loved ones, classmates, and above all to the Divine Almighty God for bestowing us with reading and peace of mind. Without their support and blessings, the completion of this study would not have been possible.

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